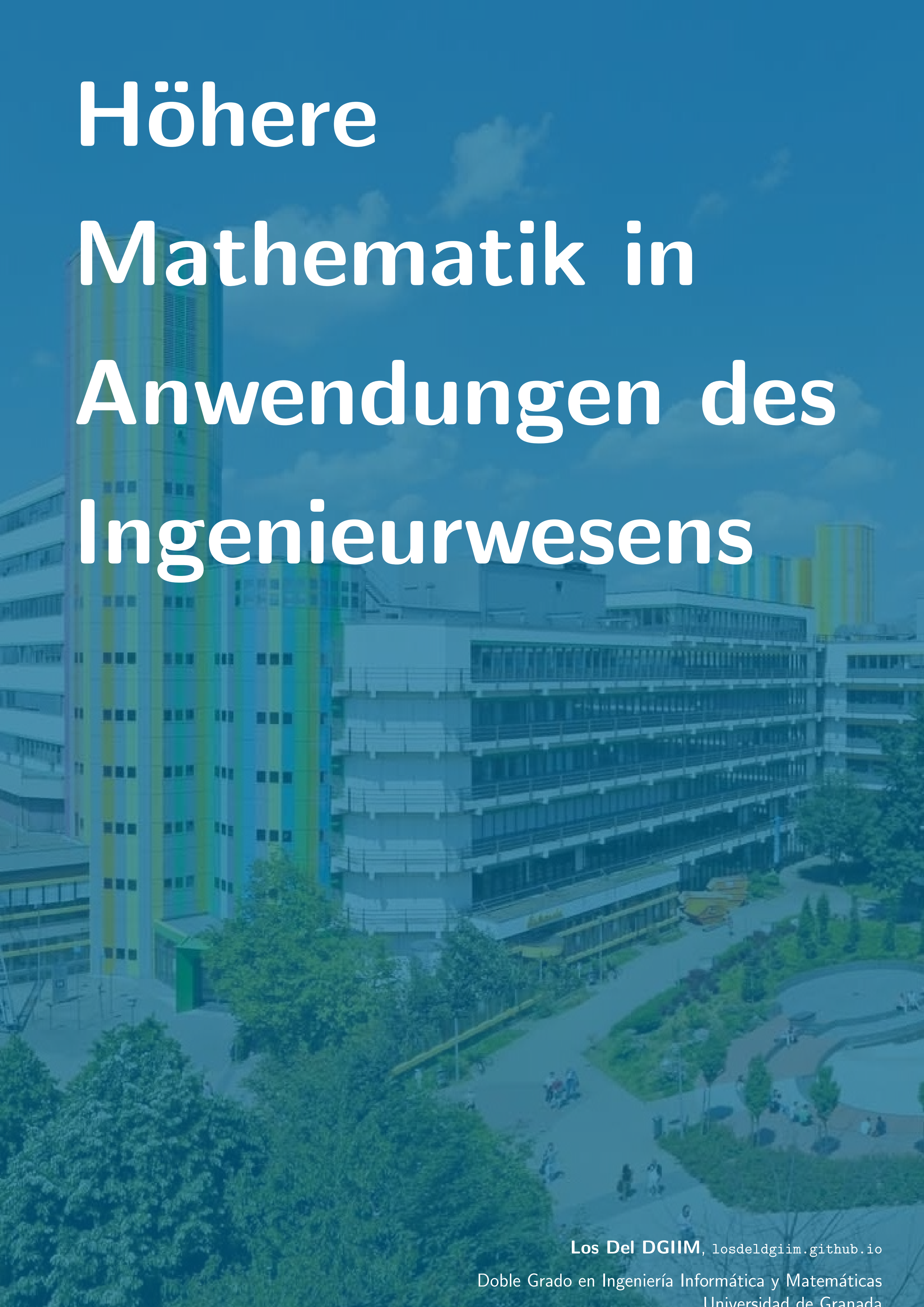

Höhere Mathematik in Anwendungen des Ingenieurwesens

Los Del DGIIM, `losdeldgiim.github.io`

Doble Grado en Ingeniería Informática y Matemáticas

Universidad de Granada

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Granada, 2025

Índice general

1. Introduction to MATLAB

MATLAB is a high-level programming language and environment used for numerical computing, data analysis, and visualization. It is widely used in engineering, science, and mathematics for tasks such as matrix manipulation, algorithm development, and data visualization. A license is required to use MATLAB, and it is commonly used in academic and research settings. However, there are alternatives to MATLAB, such as Octave and Scilab, which are open-source and free to use.

The aim of this chapter is not to teach how to use MATLAB, but rather to provide a list of useful commands and aspects that should be noted. In order to learn how to use MATLAB, it is recommended to take a course or read a tutorial on the subject. There are many resources available online, including official documentation and user forums, that can help you get started with MATLAB. Additionally, there are many books and online courses that can provide a more in-depth understanding of the language and its capabilities.

- MATLAB starts counting from 1.
- To access the last element of a vector, you can use the command `end`.
- When using vectors, the component-wise operations as `.*` for multiplication, `./` for division, and `.^` for exponentiation are used.
- Different files can be used. Variables can be kept in `.mat` files, while functions can be defined in `.m` files.
- When using a `switch` statement, if it enters a case, it will not execute the following cases, even if they are true. This is different from other programming languages where the execution continues until a `break` statement is encountered.
- Vectorization is a powerful technique in MATLAB that allows you to perform operations on entire arrays or matrices without the need for explicit loops.
- In order to count the time taken by a piece of code, you can use the `tic` and `toc` commands. The `tic` command starts a timer, and the `toc` command stops the timer and returns the elapsed time.
- Comparing vectors can have several effects depending on the context.
    - When comparing two vectors of the same size and shape, the comparison will be performed element-wise, resulting in a new vector of the same size

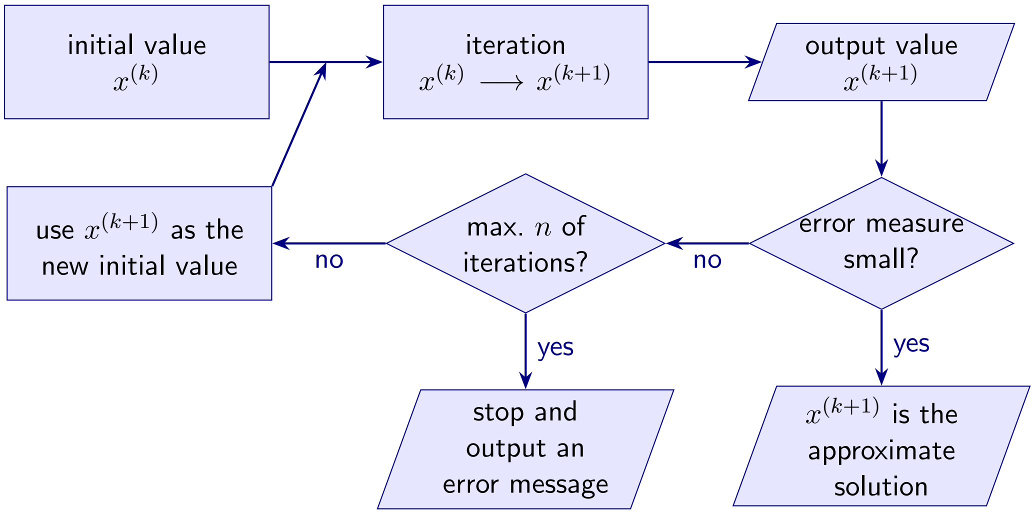

Figura 1.1: General structure of an iterative method.

and shape, where each element is the result of the comparison between the corresponding elements of the original vectors.

- When comparing a vector to a scalar, the comparison will be performed element-wise, resulting in a new vector of the same size and shape as the original vector, where each element is the result of the comparison between the corresponding element of the original vector and the scalar value.
- When comparing two vectors of the same size but opposite directions, the first vector will be compared to the second vector value by value.

1.1. Iterative Methods in MATLAB

An iterative method is a mathematical procedure that generates a sequence of approximations to the solution of a problem, with the goal of converging to the exact solution. The general structure of an iterative method can be represented as shown in Figure 1.1.

1.1.1. Floating Point Arithmetic

In order to work with numerical methods, it is important to understand how floating point arithmetic works. Floating point numbers are a way to represent real numbers in a computer with a fixed number of bits, using the representation shown in Equation 1.1.

$$x = \pm m \times 10^e \tag{1.1}$$

Where:

- m is the mantissa
- e is the exponent, which is an integer that can be positive or negative.
- The sign of the number is determined by the $\pm$ symbol.

That way, a computer can only represent a finite number of real numbers $\mathbb{M} \subset \mathbb{R}$. This leads to several issues, such as:

- Rounding errors: When $x \in \mathbb{R} \setminus \mathbb{M}$, it is approximated by a number in $\mathbb{M}$, which can lead to rounding errors.
- Catastrophic cancellation: When subtracting two nearly equal numbers, the significant digits can be lost, resulting in a large relative error.
- Overflow and underflow: When the result of a calculation exceeds the maximum representable value, it can lead to overflow, while when the result is smaller than the minimum representable value, it can lead to underflow.

  This should be taken into account when calculating the error of an iterative method, as it can affect the convergence and accuracy of the method.
- Absorbing elements: When adding a very small number to a very large number, the small number may not affect the result due to the limitations of floating point representation, leading to an absorbing element.

1.1.2. Iterative Methods for LES

In this section, we will discuss some methods that can be used to solve linear equations systems (LES). The reader may have thought of using Gaussian elimination or the inverse of the coefficient matrix, but these methods are not suitable for large systems. The MATLAB command `A\b` automatically chooses the best method iterative method to solve the system $Ax = b$. Even though iterative methods usually take longer to calculate the solution, they are ideal for:

- Sparse systems: When the coefficient matrix A has a large number of zero elements, iterative methods can take advantage of this sparsity to reduce the computational cost.
- Bad conditioned systems: Usual direct methods are really sensitive to errors in the data, while iterative methods can be more robust in these cases.

Condition Number

Given a matrix A, if A is singular (i.e., it does not have an inverse), its condition number $\text{cond}(A)$ is defined as ∞. If A is non-singular, its condition number is defined as:

$$\text{cond}(A) = \|A\| \cdot \|A^{-1}\| \geqslant \|A \cdot A^{-1}\| = \|I\| = 1$$

The condition number of a matrix is a measure of how sensitive the solution of a linear system is to changes in the input data. If $\text{cond}(A) \approx 1$, the system is well-conditioned, meaning that small changes in the input data will result in small changes in the solution. If $\text{cond}(A) \gg 1$, the system is bad-conditioned, meaning that small changes in the input data can result in large changes in the solution. The condition number can be used to calculate how good a solution is.

Proposición 1.1. *Let A be a non-singular matrix, and let x be the exact solution of the system $Ax = b$, and $\tilde{x}$ be an approximate solution. Then, the relative error of the solution can be bounded by:*

$$\frac{\|x - \tilde{x}\|}{\|x\|} \leqslant \operatorname{cond}(A) \cdot \frac{\|r\|}{\|b\|}$$

Where $r = b - A\tilde{x}$ is the residual of the approximate solution.

Demostración.

$$\begin{aligned} \|x\| \cdot \|r\| \cdot \operatorname{cond}(A) &= \|x\| \cdot \|r\| \cdot \|A\| \cdot \|A^{-1}\| \geqslant \\ &\geqslant \|Ax\| \cdot \|A^{-1}r\| = \|b\| \cdot \|A^{-1}(b - A\tilde{x})\| = \|b\| \cdot \|x - \tilde{x}\| \end{aligned}$$

Dividing both sides by $\|x\| \cdot \|b\| \geqslant 0$ gives the desired result. $\square$

2. Vectors and Matrices

This chapter is not introduced, as the reader should be familiar with it. The definitions, properties and how to calculate the following topics should be known:

- Scalar product, cross product, scalar triple product, matrix product
- Linear dependence or independence and orthogonality
- Solvability of linear systems of equations

3. Introduction to calculating with Physical Quantities

Definición 3.1 (Quantity)**.** A quantity is a property of a phenomenon, body or substance that may be distinguished qualitatively and determined quantitatively. It is expressed as the *product of a numerical value and a unit.*

There are two types of quantities:

Scalar A quantity that is completely described by its numerical value and unit. Examples include mass, time, temperature, etc.

Vectorial A quantity that is completely described by its numerical value (module), unit and direction. Examples include velocity, force, etc.

Usually, only the module of a vectorial quantity is given, and the direction is determined by the context. Other times, the direction is given by a unit vector, which is a vector of module 1 whose only purpose is to indicate the direction of the quantity.

In order to work with quantities, we need to introduce the concept of coordinate system.

3.1. Coordinate Systems

A coordinate system of $\mathbb{R}^n$ is defined by a point O called the origin and a base $\mathcal{B} = \{\overrightarrow{e}_1, \overrightarrow{e}_2, \ldots, \overrightarrow{e}_n\}$ of $\mathbb{R}^n$. The base vectors $\overrightarrow{e}_i$ are called the base vectors of the coordinate system. The coordinates of a point P in the coordinate system are the coefficients of the linear combination of the base vectors that gives the vector $\overrightarrow{OP}$:

$$\overrightarrow{OP} = \sum_{i=1}^{n} x_i \overrightarrow{e}_i$$

The basic coordinate system is the cartesian coordinate system, which is defined by the origin $O = (0, 0, 0)$ and the base $\mathcal{B}_u = \{\hat{\imath}, \hat{\jmath}, \hat{k}\}$, where $\hat{\imath}$, $\hat{\jmath}$ and $\hat{k}$ are the unit vectors in the direction of the x, y and z axes, respectively. They should always follow the right-hand rule:

$$\hat{\imath} \times \hat{\jmath} = \hat{k}$$

However, there are other coordinate systems that are more convenient to work with in certain situations. In those cases, they are usually not given by a base, but by

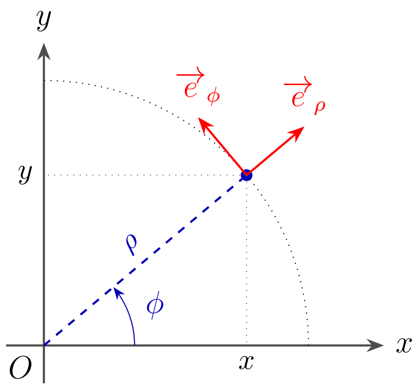

Figura 3.1: Polar coordinates.

a transformation that relates the coordinates of the new system with the coordinates of the cartesian system. In those cases, each vector of the new system is given by fixing all the coordinates of the new system except one, and letting that coordinate vary. More formally, it the new system is given by the coordinates (u_1, u_2, u_3), then the vector $\vec{e}_i$ of the new system is given by:

$$\vec{e}_i = \frac{\frac{\partial \vec{r}}{\partial u_i}}{\left\| \frac{\partial \vec{r}}{\partial u_i} \right\|}$$

where $\vec{r}$ is the position vector of a point.

3.1.1. Polar coordinates

The polar coordinate system (two dimensions) is given by the coordinates (ρ, ϕ), where ρ is the distance from the origin and ϕ is the angle between the positive x axis and the line that connects the origin with the point. It is shown in Figure 3.1. The transformation that relates the polar coordinates with the cartesian coordinates is given by:

$$\begin{cases} x = \rho \cos \phi \\ y = \rho \sin \phi \end{cases}$$

In order to calculate the base vectors of the polar coordinate system, we need to calculate the position vector of a point in polar coordinates. The position vector of a point P in polar coordinates is given by:

$$\vec{r} = x\hat{\imath} + y\hat{\jmath} = \rho \cos \phi \ \hat{\imath} + \rho \sin \phi \ \hat{\jmath}$$

Therefore:

$$\vec{e}_\rho = \frac{\cos \phi \ \hat{\imath} + \sin \phi \ \hat{\jmath}}{\sqrt{\cos^2 \phi + \sin^2 \phi}} = \cos \phi \ \hat{\imath} + \sin \phi \ \hat{\jmath}$$

$$\vec{e}_\phi = \frac{-\rho \sin \phi \ \hat{\imath} + \rho \cos \phi \ \hat{\jmath}}{\sqrt{(-\rho \sin \phi)^2 + (\rho \cos \phi)^2}} = -\sin \phi \ \hat{\imath} + \cos \phi \ \hat{\jmath}$$

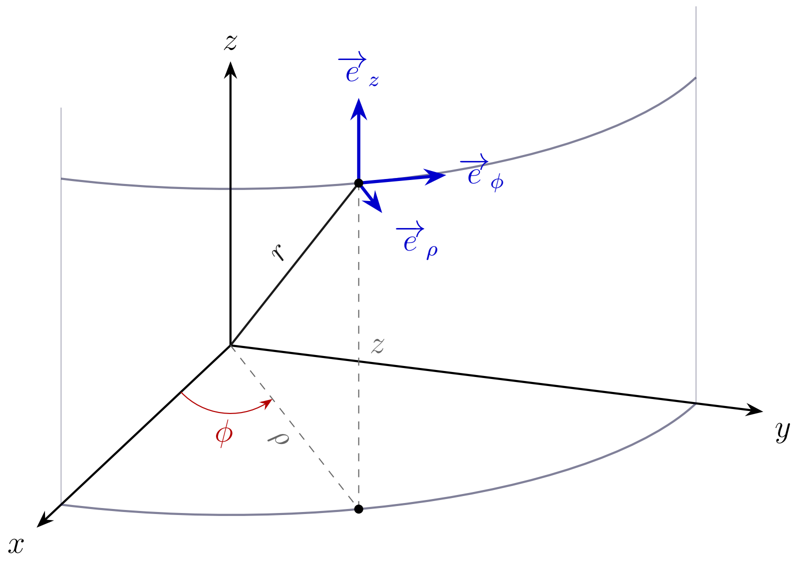

Figura 3.2: Cylindrical coordinates.

As we can see, the base vectors of the polar coordinate system are not constant, but they depend on the angle ϕ. This is because the polar coordinate system is a curvilinear coordinate system, where the axes are not straight lines, but curves. Therefore, the base vectors of a curvilinear coordinate system are not constant, but they depend on the coordinates of the point.

3.1.2. Cylindrical coordinates

The cylindrical coordinate system (three dimensions) is given by the coordinates (ρ, ϕ, z), where ρ and ϕ are the same as in the polar coordinate system when projected onto the xy plane, and z is the height above the xy plane. They are shown in Figure 3.2. The transformation that relates the cylindrical coordinates with the cartesian coordinates is given by:

$$\begin{cases} x = \rho \cos \phi \\ y = \rho \sin \phi \\ z = z \end{cases}$$

The base vectors of the cylindrical coordinate system are given by:

$$\begin{aligned} \vec{e}_\rho &= \cos \phi \ \hat{\imath} + \sin \phi \ \hat{\jmath} \\ \vec{e}_\phi &= -\sin \phi \ \hat{\imath} + \cos \phi \ \hat{\jmath} \\ \vec{e}_z &= \hat{k} \end{aligned}$$

3.1.3. Spherical coordinates

The spherical coordinate system (three dimensions) is given by the coordinates (r, θ, ϕ), where r is the distance from the origin, θ is the angle between the positive z axis and the line that connects the origin with the point, and ϕ is the angle between

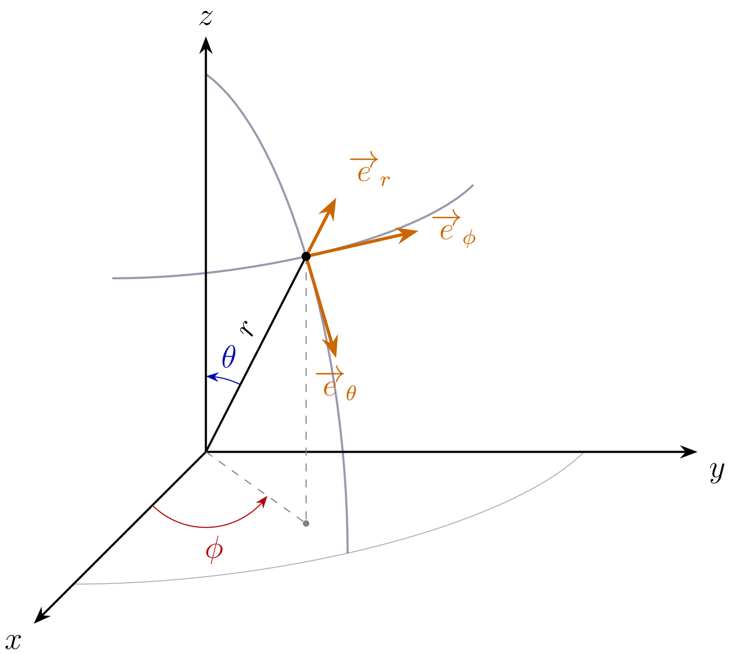

Figura 3.3: Spherical coordinates.

the positive x axis and the projection of the line that connects the origin with the point onto the xy plane. It is shown in Figure 3.3. The transformation that relates the spherical coordinates with the cartesian coordinates is given by:

$$\begin{cases} x = r\sin\theta\cos\phi \\ y = r\sin\theta\sin\phi \\ z = r\cos\theta \end{cases}$$

The base vectors of the spherical coordinate system are given by:

$$\begin{aligned} \overrightarrow{e}_r &= \sin\theta\cos\phi\ \hat{\imath} + \sin\theta\sin\phi\ \hat{\jmath} + \cos\theta\ \hat{k} \\ \overrightarrow{e}_\theta &= \cos\theta\cos\phi\ \hat{\imath} + \cos\theta\sin\phi\ \hat{\jmath} - \sin\theta\ \hat{k} \\ \overrightarrow{e}_\phi &= -\sin\phi\ \hat{\imath} + \cos\phi\ \hat{\jmath} \end{aligned}$$

4. Complex Numbers

Complex numbers are an extension of the real numbers, allowing us to work with quantities that have both a real and an imaginary part. A complex number is typically expressed in the form $z = a + bi$, where:

- $a = \Re(z)$ is the real part of the complex number.
- $b = \Im(z)$ is the imaginary part of the complex number.
- i is the imaginary unit, defined by the property $i^2 = -1$.

They can be expressed using diferent notations:

Cartesian/Algebraic Form: $z = a + bi$

Polar/Trigonometric Form: $z = r(\cos\theta + i\sin\theta)$

The idea behind the polar form is to use polar coordinates to represent the point (a, b) in the complex plane. Here, r is the magnitude (or modulus) of the complex number, and θ is the argument (or angle) of the complex number, measured from the positive real axis.

Exponential Form: $z = re^{i\theta}$

Using the Euler's formula (4.1) we can express the complex number in exponential form, which is often more convenient for multiplication and division of complex numbers.

$$e^{i\theta} = \cos\theta + i\sin\theta \tag{4.1}$$

Using the exponential form, the Moive's theorem (4.2) can be easily derived, which states that for any complex number $z = re^{i\theta}$ and any integer n:

$$z^n = r^n\left(\cos(n\theta) + i\sin(n\theta)\right) = r^n e^{in\theta} \tag{4.2}$$

The reader should be familiar with the basic operations on complex numbers and with the different representations of complex numbers, as well as with the properties of the complex plane.

5. Applications of Complex Numbers

Complex numbers have a wide range of applications in various fields of science and engineering. In the following sections, we will explore some of the most common ones.

5.1. Harmonic Oscillations

Oscillations are a fundamental phenomenon in physics, and thousands of systems can be modeled using them, starting from the simple pendulum to the vibrations of a guitar string. Harmonic oscillations are a specific type of oscillation that can be described using a sine[1] function. The general form of a harmonic oscillation can be expressed as:

$$f(t) = A\sin(\omega t + \varphi) \tag{5.1}$$

where:

- A is the amplitude of the oscillation, representing the maximum displacement from the equilibrium position.
- ω is the angular frequency, which determines how many oscillations occur in a unit of time.
- φ is the phase shift, which determines the initial position of the oscillation at time $t = 0$.
- t is the time variable.
- $f(t)$ is the value of the oscillation at time t.

In order to represent harmonic oscillations using complex numbers, the following complex function can be used, where the underline usually denotes a complex number:

$$\begin{aligned} \underline{z}(t) &= Ae^{i(\omega t+\varphi)} \\ &= A\left(\cos(\omega t + \varphi) + i\sin(\omega t + \varphi)\right) \end{aligned}$$

Therefore, the harmonic oscillation described in Equation 5.1 can be represented as $f(t) = \Im(\underline{z}(t))$, where $\Im$ denotes the imaginary part of the complex function.

[1]Cosine functions can also be used, as they are just phase-shifted versions of sine functions.

This representation allows us to analyze and manipulate harmonic oscillations just using complex numbers, which can simplify calculations and provide deeper insights into the behavior of oscillatory systems.

One additional advantage of using complex numbers is that the dependency on the time can be split. The number $\underline{z}$ can be written as:

$$\underline{z}(t) = Ae^{i\varphi}e^{i\omega t}$$

where $Ae^{i\varphi}$ is a complex constant that contains the amplitude and phase shift, while $e^{i\omega t}$ contains the time dependency. The value $Ae^{i\varphi}$ is often referred to as the *complex amplitude* $\underline{A}$, and $|\underline{A}| = A$ is the amplitude of the oscillation. This separation allows us to analyze the oscillation in terms of its amplitude and phase independently from its time evolution.

5.2. Alternating Current Circuits

Alternating current (AC) circuits are electrical circuits that operate with an alternating voltage or current. An alternating voltage can be represented as a sinusoidal function of time such as 5.2, that generates an alternating current such as 5.3:

$$v(t) = \hat{v}\sin(\omega t + \varphi_v) \tag{5.2}$$

$$i(t) = \hat{\imath}\sin(\omega t + \varphi_i) \tag{5.3}$$

where $\hat{v}$ and $\hat{\imath}$ are the peak voltage and current, ω is the angular frequency, and φ_v and φ_i are the phase shifts of the voltage and current, respectively. In AC circuits, an osciloscope can be used to visualize the voltage and current waveforms, which often appear as sinusoidal waves. If a multimeter is used to measure the voltage and current, it can't just provide the average value (as the positive and negative halves of the wave cancel each other out), but it can provide the root mean square (RMS) value, which is a measure of the effective voltage or current.

Definición 5.1 (Root Mean Square)**.** The root mean square (RMS) value of a periodic function $f(t)$ is defined as:

$$\mathrm{RMS}(f) = \sqrt{\frac{1}{T}\int_0^T f(t)^2 dt}$$

where T is the period of the function.

Proposición 5.1. *The RMS value of a sinusoidal function $f(t) = A\sin(\omega t + \varphi)$ is given by:*

$$RMS(f) = \frac{A}{\sqrt{2}}$$

Demostración.

$$\begin{aligned}
\mathrm{RMS}(f) &= \sqrt{\frac{1}{T}\int_0^T A^2\sin^2(\omega t+\varphi)dt} = A\sqrt{\frac{1}{T}\int_0^T \sin^2(\omega t+\varphi)dt} = \\
&= A\sqrt{\frac{1}{T}\int_0^T \frac{1-\cos(2\omega t+2\varphi)}{2}dt} = A\sqrt{\frac{1}{T}\left[\frac{t}{2}-\frac{\sin(2\omega t+2\varphi)}{4\omega}\right]_0^T} \overset{(*)}{=} \\
&\overset{(*)}{=} A\sqrt{\frac{1}{T}\left(\frac{T}{2}-0\right)} = A\sqrt{\frac{1}{2}} = \frac{A}{\sqrt{2}}
\end{aligned}$$

where in $(*)$ we used that T is the period of the function. □

Therefore, the RMS values read by a multimeter for the voltage (usually denoted as V) and current (I) in an AC circuit should be multiplied by $\sqrt{2}$ to obtain the peak values $\hat{v}$ and $\hat{\imath}$, respectively. As happened with harmonic oscillations, complex numbers can be used to represent the voltage and current in AC circuits. The voltage can be represented as the imaginary part of a complex function:

$$\underline{v}(t) = \hat{v}e^{j(\omega t+\varphi_v)} = \hat{v}e^{j\varphi_v}e^{j\omega t} = \underline{\hat{v}}e^{j\omega t}$$

where $\underline{\hat{v}} = \hat{v}e^{j\varphi_v}$ is the complex amplitude of the voltage. The value $\underline{V} = \sqrt{2}\underline{\hat{v}}$ is often referred to as the *complex RMS voltage*.

5.2.1. Ohm's Law and Impedance

Ohm's Law is a fundamental principle in electrical engineering that relates the voltage (V), current (I), and resistance (R) in a DC circuit:

$$V = IR \tag{5.4}$$

In AC circuits, a new concept called *impedance* ($\underline{Z}$) is introduced to generalize Ohm's Law.

Definición 5.2 (Impedance)**.** The impedance of an AC circuit is a complex quantity that relates the voltage and current in the circuit, and is defined as:

$$\underline{Z} = \frac{\underline{v}(t)}{\underline{i}(t)}$$

The general form of Ohm's Law for AC circuits can be expressed as:

$$\underline{Z} = \frac{\underline{v}(t)}{\underline{i}(t)} = \frac{\underline{\hat{v}}e^{j\omega t}}{\underline{\hat{\imath}}e^{j\omega t}} = \frac{\underline{\hat{v}}}{\underline{\hat{\imath}}} = \frac{\hat{v}e^{j\varphi_v}}{\hat{\imath}e^{j\varphi_i}} = \frac{\hat{v}}{\hat{\imath}}e^{j(\varphi_v-\varphi_i)} = \frac{V}{I}e^{j(\varphi_v-\varphi_i)}$$

As we can observe, the impedance $\underline{Z}$ is a complex number that does not depend on time. It is usually divided into its real part, called *resistance* (R), and its imaginary part, called *reactance* (X). The impedance takes into account the effects of three types of circuit elements:

- Resistors, which have a resistance R measured in ohms (Ω). The follow equation is always true for resistors:

$$v(t) = R \cdot i(t)$$

As we can see, $\underline{Z}_R = R$. In addition, the voltage and current have the same phase and angular frequency, but the amplitude is different.

- Inductors, which have an inductance L measured in henries (H). The follow equation is always true for inductors:

$$\underline{v}(t) = L\frac{d\underline{i}(t)}{dt}$$

Operating the derivative, we can obtain:

$$\underline{v}(t) = L\frac{d\underline{i}(t)}{dt} = L\frac{d}{dt}\left(\hat{\underline{i}}e^{j\omega t}\right) = j\omega L\hat{\underline{i}}e^{j\omega t} = j\omega L\underline{i}(t) \Longrightarrow \underline{Z}_L = j\omega L$$

Taking into account the definition of impedance, we can conclude that $\underline{Z}_L = j\omega L$. In addition, the voltage and current have the same angular frequency, but there is a phase shift of $-\pi/2$ between them, and the amplitude is different.

- Capacitors, which have a capacitance C measured in farads (F).

  The follow equation is always true for capacitors:

$$\underline{i}(t) = C\frac{d\underline{v}(t)}{dt}$$

Operating the derivative, we can obtain:

$$\underline{i}(t) = C\frac{d\underline{v}(t)}{dt} = C\frac{d}{dt}\left(\hat{\underline{v}}e^{j\omega t}\right) = j\omega C\hat{\underline{v}}e^{j\omega t} = j\omega C\underline{v}(t) \Longrightarrow \underline{Z}_C = \frac{1}{j\omega C}$$

Taking into account the definition of impedance, we can conclude that $\underline{Z}_C = \frac{1}{j\omega C}$. In addition, the voltage and current have the same angular frequency, but there is a phase shift of $\pi/2$ between them, and the amplitude is different.

"Solving" an AC circuit means calculating the voltage, current, and impedance in the circuit. In order to do so, first of all the equivalent impedance of the circuit must be calculated using the rules for series and parallel connections of impedances.

- Serial connections of impedances are impedences that are connected end-to-end, so the same current flows through each impedance. The equivalent impedance of a series connection is given by the sum of the individual impedances:

$$\underline{Z}_{\text{series}} = \sum_{k=1}^{n} \underline{Z}_k$$

- Parallel connections of impedances are impedences that are connected across the same two points, so the same voltage is applied across each impedance. The equivalent impedance of a parallel connection is given by the reciprocal of the sum of the reciprocals of the individual impedances:

$$\frac{1}{\underline{Z}_{\text{parallel}}} = \sum_{k=1}^{n} \frac{1}{\underline{Z}_k}$$

After that, and appart from the Ohm's Law, the Kirchhoff's circuit laws can be used to solve the circuit.

- Kirchhoff's Current Law (KCL) states that the total current entering a junction in a circuit must equal the total current leaving the junction. Mathematically, it can be expressed as:

$$\sum_{k=1}^{n} I_k = 0$$

  where I_k is each current entering or leaving the junction, with currents entering the junction considered positive and currents leaving the junction considered negative.

- Kirchhoff's Voltage Law (KVL) states that the total voltage around any closed loop in a circuit must equal zero. Mathematically, it can be expressed as:

$$\sum_{k=1}^{n} V_k = 0$$

  where V_k is the voltage difference across each element in the loop, following a consistent direction (e.g., clockwise or counterclockwise).

6. Analysis of Functions of One Variable

This chapter is not introduced, as the reader should be familiar with it. Regarding derivation, the following topics should be known:

- Derivation of polynomials, root functions, trigonometric functions, logarithms and exponential functions,
- Product rule, quotient rule, chain rule,
- Derivation of a function elevated to another function,

$$\frac{d}{dx}\left(f(x)^{g(x)}\right) = \frac{d}{dx}\left(e^{\ln\left(f(x)^{g(x)}\right)}\right) = \frac{d}{dx}\left(e^{g(x)\cdot\ln(f(x))}\right) = e^{g(x)\cdot\ln(f(x))}\cdot\frac{d}{dx}\left(g(x)\cdot\ln\left(f(x)\right)\right) =$$
$$= f(x)^{g(x)}\cdot\left(g'(x)\cdot\ln\left(f(x)\right) + g(x)\cdot\frac{f'(x)}{f(x)}\right).$$

- Curve discussion.

Regarding integration, the following topics should be known:

- Integration of polynomials, root functions, trigonometric functions, logarithms and exponential functions,
- Integration by parts, substitution rule,

7. Analysis of Functions of More than One Variable

Functions of more than one variable are common in many fields, including physics, engineering, and economics. In this chapter, we will explore the analysis of such functions.

7.1. Functional Limits

In order to introduce the concept of differentiability for functions of more than one variable, we first need to define some previous concepts such as functional limits. This concept requires however some topological concepts such as neighborhood, open set, boundary point, boundary, closure, and convergence. Even though these terms could have a much general definition in term of topological spaces, but we will stick to the Euclidean space $\mathbb{R}^n$, and we will give the definitions in this context.

Definición 7.1 (Neighborhood)**.** Given a point $x \in \mathbb{R}^n$, a neighborhood of x is, for each $\varepsilon > 0$, the set

$$N_\varepsilon(x) = \{y \in \mathbb{R}^n : \|y - x\| < \varepsilon\}.$$

Definición 7.2 (Open Set)**.** A set $U \subseteq \mathbb{R}^n$ is called open if for every point $x \in U$, there exists an $\varepsilon > 0$ such that $N_\varepsilon(x) \subseteq U$.

Proposición 7.1. *Given a collection of open sets $\{U_i\}_{i\in I}$:*

- *The union $\bigcup\limits_{i\in I} U_i$ is open.*
- *The intersection $\bigcap\limits_{i=1}^{n} U_i$ of a finite number of open sets is open.*

Definición 7.3 (Complement)**.** Given a set $A \subseteq \mathbb{R}^n$, the complement of A is the set

$$A^c := \mathbb{R}^n \setminus A = \{x \in \mathbb{R}^n : x \notin A\}.$$

Definición 7.4 (Boundary Point)**.** Given a set $A \subseteq \mathbb{R}^n$, a point $x \in \mathbb{R}^n$ is called a boundary point of A if:

$$\forall \varepsilon > 0, \qquad N_\varepsilon(x) \cap A \neq \emptyset \quad \wedge \quad N_\varepsilon(x) \cap A^c \neq \emptyset.$$

Definición 7.5 (Boundary)**.** Given a set $A \subseteq \mathbb{R}^n$, the boundary of A, denoted by ∂A, is the set of all boundary points of A.

Definición 7.6 (Closure)**.** Given a set $A \subseteq \mathbb{R}^n$, the closure of A, denoted by $\overline{A}$, is the set

$$\overline{A} := A \cup \partial A.$$

Definición 7.7 (Convergence)**.** A sequence $\{x_n\}_{n\in\mathbb{N}}$ in $\mathbb{R}^n$ is said to converge to a point $x \in \mathbb{R}^n$ if:

$$\forall \varepsilon > 0, \qquad \exists M \in \mathbb{N} : n > M \implies x_n \in N_\varepsilon(x).$$

In this case, we write $\lim\limits_{n\to\infty} x_n = x$ or $\{x_n\} \to x$ as $n \to \infty$.

Proposición 7.2. *A sequence $\{x_n\}_{n\in\mathbb{N}}$ in $\mathbb{R}^n$ converges to a point $x \in \mathbb{R}^n$ each component converges to the corresponding component of x. In other words, these two statements are equivalent:*

- $\{x_n\} \to x$ *as* $n \to \infty$.
- $\{x_i^n\} \to x_i$ *as* $n \to \infty$ *for each* $i = 1, 2, \ldots, n$.

Once we have defined the concept of convergence, we can define the concept of functional limits.

Definición 7.8 (Functional Limit)**.** Let $G \subset \mathbb{R}^n$ be a non-empty set, and let $f : G \to \mathbb{R}$ be a function. We say that the limit of f as x approaches x_0 exists and is equal to L if, for every sequence $\{x_n\}_{n\in\mathbb{N}}$ in G that converges to x_0, the sequence $\{f(x_n)\}_{n\in\mathbb{N}}$ converges to L. This is:

$$\forall \{x_n\} \subseteq G, \qquad \{x_n\} \to x_0 \implies \{f(x_n)\} \to L.$$

In this case, we write $\lim\limits_{x\to x_0} f(x) = L$.

7.2. Continuity

Continuity is a fundamental concept in the analysis of functions. It allows us to understand how a function behaves near a point and is essential for defining differentiability.

Definición 7.9 (Continuity)**.** Let $G \subset \mathbb{R}^n$ be a non-empty set, and let $f : G \to \mathbb{R}$ be a function. We say that f is continuous at a point $x_0 \in G$ if the limit of f as x approaches x_0 exists and is equal to $f(x_0)$. This is:

$$\lim_{x\to x_0} f(x) = f(x_0).$$

We say that f is continuous on $H \subset G$ if f is continuous at every point of H.

Proposición 7.3. *Let $G \subset \mathbb{R}^n$ be a non-empty set, and let $f : G \to \mathbb{R}$ be a function. The following statements are equivalent:*

- f *is continuous at a point* $x_0 \in G$.
- f_i *is continuous at* x_0 *for each* $i = 1, 2, \ldots, n$.

7.3. Differentiability

Differentiability is a stronger condition than continuity. It allows us to understand how a function changes near a point and is essential for defining the concept of the derivative. Previously, we have to define some last concepts.

Definición 7.10 (Connected Set)**.** A set $A \subseteq \mathbb{R}^n$ is called connected if, for any two points $x, y \in A$, there exists a polygonal path that connects x and y and is entirely contained in A.

Definición 7.11 (Domain)**.** A non-empty set $G \subseteq \mathbb{R}^n$ is called a domain if it is open and connected.

Teorema 7.4 (Intermediate Value Theorem)**.** *Let $G \subset \mathbb{R}^n$ be a connected set, and let $f : G \to \mathbb{R}$ be a continuous function. Consider two points $x, y \in G$ such that $f(x) < f(y)$. Then, for every $c \in [f(x), f(y)]$, there exists a point $z \in G$ such that $f(z) = c$.*

Definición 7.12 (Bounded Set)**.** A set $A \subseteq \mathbb{R}^n$ is called bounded if there exists a positive real number M such that $\|x\| < M$ for all $x \in A$.

Teorema 7.5 (Extreme Value Theorem)**.** *Let $G \subset \mathbb{R}^n$ be a closed and bounded set, and let $f : G \to \mathbb{R}$ be a continuous function. Then, there exist points $x_{\text{mín}}, x_{\text{máx}} \in G$ such that*

$$f(x_{\text{mín}}) \leqslant f(x) \leqslant f(x_{\text{máx}}) \quad \text{for all } x \in G.$$

Once those concepts are defined, we can define the concept of differentiability.

Definición 7.13 (Differentiability)**.** Let $G \subset \mathbb{R}^n$ be a domain, and let $f : G \to \mathbb{R}^m$ be a function. We say that f is differentiable at a point $x_0 \in G$ if there exists a linear transformation $L : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{x \to x_0} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0.$$

In this case, L is unique and we write $Df(x_0) = L$ and call L the derivative of f at x_0.

We say that f is differentiable on $H \subset G$ if f is differentiable at every point of H. We then call $Df : H \to L(\mathbb{R}^n, \mathbb{R}^m)$ the derivative of f on H.

Some properties of differentiable functions are the following:

Proposición 7.6. *Let $G \subset \mathbb{R}^n$ be a domain, and let $f : G \to \mathbb{R}^m$ be a function. If f is differentiable at a point $x_0 \in G$, then f is continuous at x_0.*

Proposición 7.7. *Let $G \subset \mathbb{R}^n$ be a domain, and let $f : G \to \mathbb{R}^m$ be a function. For every $\alpha \in \mathbb{R}$, if f is differentiable at a point $x_0 \in G$, then αf is differentiable at x_0 and:*

$$D(\alpha f)(x_0) = \alpha Df(x_0).$$

Proposición 7.8. *Let $G \subset \mathbb{R}^n$ be a domain, and let $f, g : G \to \mathbb{R}^m$ be functions. If f and g are differentiable at a point $x_0 \in G$, then $f + g$ is differentiable at x_0 and:*

$$D(f + g)(x_0) = Df(x_0) + Dg(x_0).$$

Proposición 7.9 (Chain Rule)**.** *Let $G \subset \mathbb{R}^n$ be a domain, and let $f : G \to \mathbb{R}^m$. Let $g : f(G) \to \mathbb{R}^r$ be a function. If f is differentiable at a point $x_0 \in G$, and g is differentiable at the point $f(x_0)$, then the composition $g \circ f : G \to \mathbb{R}^r$ is differentiable at x_0, and:*

$$D(g \circ f)(x_0) = Dg(f(x_0)) \circ Df(x_0).$$

Proposición 7.10. *Let $G \subset \mathbb{R}^n$ be a domain, and let $f, g : G \to \mathbb{R}^m$ be functions. If f and g are differentiable at a point $x_0 \in G$, then the dot product $f \cdot g$ is differentiable at x_0 and:*

$$D(f \cdot g)(x_0) = g(x_0) \cdot Df(x_0) + f(x_0) \cdot Dg(x_0).$$

Proposición 7.11. *Let $G \subset \mathbb{R}^n$ be a domain, and let $f : G \to \mathbb{R}^m$ and $g : G \to \mathbb{R}$ be functions. If f and g are differentiable at a point $x_0 \in G$ such that $g(x_0) \neq 0$, then the division ${}^f\!/_g$ is differentiable at x_0 and:*

$$D\left(\frac{f}{g}\right)(x_0) = \frac{g(x_0) \cdot Df(x_0) - f(x_0) \cdot Dg(x_0)}{g(x_0)^2}.$$

7.3.1. Directional Derivatives

Let $G \subset \mathbb{R}^n$ be a domain, and let $f : G \to \mathbb{R}^m$ be a function. Given a point $x_0 \in G$ and a vector $v \in \mathbb{R}^n$ such that $\|v\| = 1$, we say that f is directionally differentiable at x_0 in the direction of v if exists a linear transformation $L : \mathbb{R}^n \to \mathbb{R}^m$ such that:

$$\lim_{t \to 0} \frac{\|f(x_0 + tv) - f(x_0) - L(tv)\|}{|t|} = 0.$$

The vector $L(v)$ is called the directional derivative of f at x_0 in the direction of v, and we denote it by:

$$\frac{\partial f}{\partial v}(x_0) = L(v)$$

Observación. It should be noted that the definition of directional derivative is similar to the definition of differentiability, but instead of considering the limit as x approaches x_0 in any direction, we only consider when x approaches x_0 along the direction of the vector v, i.e., when $x = x_0 + tv$ for some scalar t that approaches 0. Therefore, if a function is differentiable at a point, then it is also directionally differentiable at that point in every direction.

In the practice, the directional derivative is just a normal derivative of a single variable function. In fact, if we define the function:

$$\begin{array}{rccl} \varphi_v : &]-\delta, \delta[& \longrightarrow & \mathbb{R}^m \\ & t & \longmapsto & f(x_0 + tv) \end{array}$$

where $\delta > 0$ is such that $x_0 + tv \in G$ for all $t \in]-\delta, \delta[$, then the directional derivative of f at x_0 in the direction of v is just the derivative of φ_v at $t = 0$, i.e.:

$$\frac{\partial f}{\partial v}(x_0) = \varphi_v'(0).$$

where, if $m > 1$, the derivative $\varphi_v'(0)$ is just the vector of the derivatives of each component of φ_v at $t = 0$.

7.3.2. Partial Derivatives

Let $G \subset \mathbb{R}^n$ be a domain, and let $f : G \to \mathbb{R}^m$ be a function. Given a point $x_0 \in G$ and an index $i \in \{1, 2, \ldots, n\}$, we say that f is partially differentiable at x_0 with respect to the variable x_i if it is directionally differentiable at x_0 in the direction of the standard basis vector e_i, where e_i is the vector with a 1 in the i-th position and 0 in all other positions. The partial derivative of f at x_0 with respect to the variable x_i is denoted by:

$$\frac{\partial f}{\partial x_i}(x_0) = \frac{\partial f}{\partial e_i}(x_0).$$

7.3.3. Jacobian Matrix

Let $G \subset \mathbb{R}^n$ be a domain, and let $f : G \to \mathbb{R}^m$ be a function. If f is differentiable at a point $x_0 \in G$, then the derivative $Df(x_0)$, as a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$, can be represented by a matrix. This matrix is called the Jacobian matrix of f at x_0, and it is denoted by $Jf(x_0) \in \mathcal{M}_{m\times n}(\mathbb{R})$.

Proposición 7.12. *Let $G \subset \mathbb{R}^n$ be a domain, and let $f : G \to \mathbb{R}^m$ be a function. If f is differentiable at a point $x_0 \in G$, then the Jacobian matrix of f at x_0 is given by:*

$$Jf(x_0) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(x_0) & \frac{\partial f_1}{\partial x_2}(x_0) & \cdots & \frac{\partial f_1}{\partial x_n}(x_0) \\ \frac{\partial f_2}{\partial x_1}(x_0) & \frac{\partial f_2}{\partial x_2}(x_0) & \cdots & \frac{\partial f_2}{\partial x_n}(x_0) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(x_0) & \frac{\partial f_m}{\partial x_2}(x_0) & \cdots & \frac{\partial f_m}{\partial x_n}(x_0) \end{pmatrix} = \left(\frac{\partial f_i}{\partial x_j}(x_0)\right)_{1\leqslant i\leqslant m, 1\leqslant j\leqslant n}.$$

If $n = 1$, i.e., if $G \subset \mathbb{R}$, then the Jacobian matrix of f at x_0 is just a column vector, and it is given by:

$$Jf(x_0) = \begin{pmatrix} f_1'(x_0) \\ f_2'(x_0) \\ \vdots \\ f_m'(x_0) \end{pmatrix}$$

where $f_i'(x_0)$ is the derivative of the i-th component of f at x_0. We call this the derivative vector of f at x_0.

If $m = 1$, i.e., if $f : G \to \mathbb{R}$ is a scalar-valued function, then the Jacobian matrix of f at x_0 is just a row vector, and it is given by:

$$Jf(x_0) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(x_0) & \frac{\partial f}{\partial x_2}(x_0) & \cdots & \frac{\partial f}{\partial x_n}(x_0) \end{pmatrix}$$

This is the gradient of f at x_0, and we denote it by $\nabla f(x_0) = \text{grad } f(x_0) = Jf(x_0)$. It should be noted that the gradient of a scalar-valued function is a row vector that points in the direction of the greatest rate of increase of the function, and its magnitude represents the rate of increase in that direction.

7.3.4. Total Differentiability

The total differentiability of a function lets us understand how a does the value of a function change when we change all the variables at the same time. The term dx represents a small change in the input variables, and the term df represents the corresponding change in the output of the function. Considering G a domain, and $f : G \to \mathbb{R}^m$ a function, its total differential at a point $x_0 \in G$ is given by:

$$df = Jf(x_0) \cdot dx$$

f scalar-valued functions, i.e., when $m = 1$, where the total differential is given by:

$$df = \nabla f(x_0) \cdot dx = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(x_0) dx_i$$

7.4. Integration of Functions of More than One Variable

Integration of functions of more than one variable is a generalization of the concept of integration for functions of a single variable. It allows us to compute the volume under a surface, the mass of an object, and many other quantities that depend on multiple variables.

7.4.1. Curve Integrals

Let $[a, b]$ be a closed interval in $\mathbb{R}$, and let $\varphi : [a, b] \to \mathbb{R}^n$ be a continuous function. We say that φ is the parametrization of a curve $C \subset \mathbb{R}^n$ given by:

$$C = \{\varphi(t) : t \in [a, b]\}.$$

Observación. It should be noted that the curve C is the image of the parametrization φ, and it is a subset of $\mathbb{R}^n$. A curve may have multiple parametrizations. For example, the curve C given by the unit circle in $\mathbb{R}^2$ can be parametrized by $\varphi(t) = (\cos t, \sin t)$ for $t \in [0, 2\pi]$, or by $\psi(t) = (\cos(2t), \sin(2t))$ for $t \in [0, \pi]$. Both φ and ψ represent the same curve C, which is the unit circle.

The parameterization φ is said to be *regular* if φ is differentiable and $\varphi'(t) \neq 0$ for all $t \in \,]a, b[$. In this case, we say that C is a regular curve.

Observación. A parameterization φ can be seen as how we traverse the curve C. The parameter t can be thought of as time, and $\varphi(t)$ gives us the position on the curve at time t. The vector $\varphi'(t)$ represents the velocity of the traversal at time t. If $\varphi'(t) \neq 0$ for all t, it means that we are always moving along the curve and never stopping, which is why we call it a regular parameterization. If $\varphi'(t) = 0$ for some t, it means that we are momentarily stopping at that point on the curve, which can lead to complications (as the direction could change abruptly), and that's why we don't consider it regular.

Length of a Curve

The length of a curve C is given by the supremum of the lengths of polygonal paths inscribed in C. If that supremum is finite, we say that C is rectifiable, and we denote its length by $\ell(C)$.

Proposición 7.13. *Let $[a,b]$ be a closed interval in $\mathbb{R}$, and let $\varphi : [a,b] \to \mathbb{R}^n$ be a regular parameterization of a curve C. Then, C is rectifiable. If φ is injective, then:*

$$\ell(C) = \int_a^b \|\varphi'(t)\| dt.$$

Curve Integral of First Kind

Let G be a *domain* in $\mathbb{R}^n$ (i.e., an open and connected set), and let $f : G \to \mathbb{R}$ be a *continuous* function. Let C be a curve in G with a *regular* parameterization $\varphi : [a,b] \to G$. The curve integral of first kind of f along φ is defined as:

$$\int_\varphi f = \int_a^b f(\varphi(t))\|\varphi'(t)\| dt.$$

Observación. The curve integral of first kind can be thought of a "weighted" length of the curve C, where the weight at each point $\varphi(t)$ is given by the value of the function f at that point. If f is constant, then the curve integral of first kind reduces to the length of the curve multiplied by that constant.

Proposición 7.14 (Linearity of the Curve Integral of First Kind)**.** *Let G be a domain in $\mathbb{R}^n$, and let $f, g : G \to \mathbb{R}$ be continuous functions. Let C be a curve in G with a regular parameterization $\varphi : [a,b] \to G$. Then:*

- *For every $\alpha \in \mathbb{R}$, we have* $\displaystyle\int_\varphi \alpha f = \alpha \int_\varphi f$.
- $\displaystyle\int_\varphi (f+g) = \int_\varphi f + \int_\varphi g$.

Proposición 7.15 (Orientation of the Curve Integral of First Kind)**.** *Let G be a domain in $\mathbb{R}^n$, and let $f : G \to \mathbb{R}$ be a continuous function. Let C be a curve in G with a regular parameterization $\varphi : [a,b] \to G$. If we reverse the orientation of the curve by considering the parameterization $\psi : [a,b] \to G$ given by $\psi(t) = \varphi(a+b-t)$, then:*

$$\int_\psi f = \int_\varphi f.$$

Proposición 7.16 (Modulus of the Curve Integral of First Kind)**.** *Let G be a domain in $\mathbb{R}^n$, and let $f : G \to \mathbb{R}$ be a continuous function. Let C be a curve in G with a regular parameterization $\varphi : [a,b] \to G$. Then:*

$$\left| \int_\varphi f \right| \leqslant \int_\varphi |f|.$$

Some important applications of the curve integral of first kind are the following:

- Let's suppose that we have a wire in the shape of a curve C in $\mathbb{R}^3$, and we want to compute the mass of that wire. If we know the density of the wire at each point, which is given by a continuous function $f : C \to \mathbb{R}$, then the mass of the wire can be computed as the curve integral of first kind of f along a regular parameterization of C. That is:

$$\text{Mass} = \int_\varphi f.$$

- In order to calculate "fence areas" we can use the curve integral of first kind. Given a domain G in $\mathbb{R}^2$, a curve C in G with a regular inyective parameterization $\varphi : [a, b] \to G$, and a continuous function $f : G \to \mathbb{R}$ such that it represents the height of a fence at each point of G, then the area of the fence can be computed as the curve integral of first kind of f along φ. That is:

$$\text{Area} = \int_\varphi f.$$

  Mathematically, we compute the area of:

$$\{(\varphi_1(t), \varphi_2(t), z) : z \in [0, f(\varphi(t))],\ t \in [a, b]\}$$

Curve Integral of Second Kind

Let G be a *domain* in $\mathbb{R}^n$, and let $f : G \to \mathbb{R}^n$ be a *continuous* function. Let C be a curve in G with a *regular* parameterization $\varphi : [a, b] \to G$. The curve integral of second kind of f along φ is defined as:

$$\int_\varphi f = \int_a^b (f(\varphi(t)) \cdot \varphi'(t))\, dt.$$

Once important aspect to take into account is the tangential component of the vector field f along the parameterization φ. The tangential component of f along φ at a point $\varphi(t)$ is given by the projection of $f(\varphi(t))$ onto the direction of $\varphi'(t)$, which is given by:

$$\text{proj}_{\varphi'(t)} f(\varphi(t)) = \|f(\varphi(t))\| \cos(\alpha(t))$$

where $\alpha(t)$ is the angle between the vectors $f(\varphi(t))$ and $\varphi'(t)$. Therefore, the curve integral of second kind can be thought of as the integral of the tangential component of f along the parameterization φ.

Proposición 7.17 (Linearity of the Curve Integral of Second Kind)**.** *Let G be a domain in $\mathbb{R}^n$, and let $f, g : G \to \mathbb{R}^n$ be continuous functions. Let C be a curve in G with a regular parameterization $\varphi : [a, b] \to G$. Then:*

- *For every $\alpha \in \mathbb{R}$, we have* $\displaystyle\int_\varphi \alpha f = \alpha \int_\varphi f$.
- $\displaystyle\int_\varphi (f + g) = \int_\varphi f + \int_\varphi g$.

Proposición 7.18 (Orientation of the Curve Integral of Second Kind)**.** *Let G be a domain in $\mathbb{R}^n$, and let $f : G \to \mathbb{R}^n$ be a continuous function. Let C be a curve in G with a regular parameterization $\varphi : [a, b] \to G$. If we reverse the orientation of the curve by considering the parameterization $\psi : [a, b] \to G$ given by $\psi(t) = \varphi(a+b-t)$, then:*

$$\int_\psi f = -\int_\varphi f.$$

Proposición 7.19 (Modulus of the Curve Integral of Second Kind)**.** *Let G be a domain in $\mathbb{R}^n$, and let $f : G \to \mathbb{R}^n$ be a continuous function. Let C be a curve in G with a regular parameterization $\varphi : [a, b] \to G$. Then:*

$$\left| \int_\varphi f \right| \leqslant \int_\varphi \|f\|.$$

Some important applications of the curve integral of second kind are the following:

- In order to calculate the area enclosed by a curve C in $\mathbb{R}^2$, we can use the curve integral of second kind. Given a domain G in $\mathbb{R}^2$, a curve C in G with a regular parameterization $\varphi : [a, b] \to G$ that traverses the curve only once in a counterclockwise direction, the area enclosed by the curve C can be computed as:

  $$\text{Area} = \frac{1}{2} \int_\varphi (-y, x)$$

  - If enclosing the area in a clockwise direction needs k regular parameterizations $\varphi_1, \varphi_2, \dots, \varphi_k$, then the area can be computed as:

    $$\text{Area} = \frac{1}{2} \sum_{i=1}^{k} \int_{\varphi_i} (-y, x).$$

- In order to calculate the work done by a force field $f : G \to \mathbb{R}^n$ along a curve C in G, we can use the curve integral of second kind. Given a domain G in $\mathbb{R}^n$, a curve C in G with a regular parameterization $\varphi : [a, b] \to G$, and a continuous function $F : G \to \mathbb{R}^n$ that represents the force field, then the work done by the force field along the curve can be computed as:

  $$W = \int_\varphi F.$$

7.4.2. Potential fields

Given a domain G in $\mathbb{R}^n$, and a continuous vector field $F : G \to \mathbb{R}^n$, we say that F is a potential field if there exists a differentiable scalar-valued function $\varphi : G \to \mathbb{R}$ such that:

$$F = \nabla\varphi = \left(\frac{\partial \varphi}{\partial x_1}, \frac{\partial \varphi}{\partial x_2}, \dots, \frac{\partial \varphi}{\partial x_n} \right).$$

In this case, we say that φ is a potential function for the potential field F. For instance, the electric field generated by a point charge is a potential field, and its potential function is the electric potential:

$$\overrightarrow{E} = -\nabla V$$

Potential fields have the following important property:

Proposición 7.20. *Let G be a domain in $\mathbb{R}^n$, and let $F : G \to \mathbb{R}^n$ be a continuous vector field. If F is a potential field with potential function $\varphi : G \to \mathbb{R}$, then for every curve C in G with a regular parameterization $\psi : [a, b] \to G$, we have:*

$$\int_\psi F = \varphi(\psi(b)) - \varphi(\psi(a)).$$

That is, the curve integral of second kind of a potential field along a curve C depends only on the endpoints of the curve, and not on the particular path taken between those endpoints. In order to find a potential function for a given vector field, we can use the following proposition.

Proposición 7.21. *Let G be a domain in $\mathbb{R}^n$, and let $F : G \to \mathbb{R}^n$ be a continuous vector field. If a curve integral of second kind of F along any curve C in G depends only on the endpoints of the curve, then F is a potential field. That is, for every two points $x_1, x_2 \in G$ and every regular parameterization $\psi : [a, b] \to G$ such that $\psi(a) = x_1$ and $\psi(b) = x_2$, the following integral depends only on the endpoints x_1 and x_2, i.e.:*

$$\int_\psi F$$

depends only on x_1 and x_2 and not on the particular parameterization ψ.

Even though this proposition provides us a theoretical way to prove that a vector field is in fact a potential field, in practice determining if a curve integral is path independent or not is not so easy. One condition that every potential field must satisfy is the following.

Proposición 7.22. *Let G be a domain in $\mathbb{R}^n$, and let $F : G \to \mathbb{R}^n$ be a continuous vector field. If F is a potential field with potential function $\varphi : G \to \mathbb{R}$, then the partial derivatives of φ satisfy the following conditions:*

$$\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i} \quad \text{for all } i, j \in \{1, 2, \dots, n\}.$$

This condition is necessary for a vector field to be a potential field, but it is not sufficient. In order to apply it in the opposite direction, we need to add some extra conditions on the domain G: it has to be simply connected, which means that it has no holes. We will however not go into the details of simply connected domains, as star-shaped domains are simply connected.

Definición 7.14 (Star-shaped)**.** A set $G \subset \mathbb{R}^n$ is called star-shaped from a point $x_0 \in G$ if for every point $x \in G$, the line segment connecting x_0 and x is entirely contained in G. That is, for every $x \in G$ and every $t \in [0, 1]$, we have:

$$(1 - t)x_0 + tx \in G.$$

We say that G is a star-shaped domain if there exists a point $x_0 \in G$ such that G is star-shaped from x_0.

Definición 7.15 (Convex)**.** A set $G \subset \mathbb{R}^n$ is called convex if for every two points $x, y \in G$, the line segment connecting x and y is entirely contained in G. That is, for every $x, y \in G$ and every $t \in [0, 1]$, we have:

$$(1 - t)x + ty \in G.$$

It is easy to see that every convex set is star-shaped, which will be useful for us, as we will see in the next proposition.

Proposición 7.23. *Let G be a star-shaped domain in $\mathbb{R}^n$, and let $F : G \to \mathbb{R}^n$ be a continuous vector field. If the partial derivatives of F satisfy the following conditions:*

$$\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i} \quad \text{for all } i, j \in \{1, 2, \dots, n\},$$

then F is a potential field.

This is a very useful result, as it provides us a way to determine if a vector field is a potential field or not by just checking the equality of the mixed partial derivatives. In practice, this is much easier than trying to determine if a curve integral is path independent or not. Once we have proved that a vector field is a potential field, there are two main ways to find a potential function for that vector field:

1. An origin point x_0 is chosen in the domain G to serve as a reference point. For any point x in the domain, a curve ψ is defined that connects x_0 to x:

   $$\begin{array}{rccl} \psi : & [0, 1] & \longrightarrow & G \\ & t & \longmapsto & x_0 + t(x - x_0) \end{array}$$

   The potential function φ is then defined as the curve integral of second kind of the vector field F along the curve ψ:

   $$\varphi(x) = \int_\psi F.$$

   Ideally, the origin point x_0 could be chosen as the origin of the coordinate system, i.e., $x_0 = (0, 0, \dots, 0)$, so that the curve ψ would be even simpler.

2. The potential function φ can be directly computed by integrating the components of the vector field F.

7.4.3. Multiple Integrals

In this section, we will introduce the concept of multiple integrals, which are a generalization of the concept of integration for functions of a single variable. Multiple integrals allow us to compute volumes, masses, and other quantities that depend on multiple variables. In order to introduce them, normal areas need to be defined.

Definición 7.16 (Normal Area)**.** A normal area in $\mathbb{R}^n$ is inductively defined as follows:

- A normal area in $\mathbb{R}$ is just a closed interval $[a, b]$.
- A normal area in $\mathbb{R}^n$ for $n > 1$ is a set of the form:

  $$A = \{(x_1, \dots, x_{n-1}, x_n) \in \mathbb{R}^n : (x_1, \dots, x_{n-1}) \in A',\ g_1(x_1, \dots, x_{n-1}) \leqslant x_n \leqslant g_2(x_1, \dots, x_{n-1})\}$$

  where A' is a normal area in $\mathbb{R}^{n-1}$, and $g_1, g_2 : A' \to \mathbb{R}$ are continuous functions such that:

  $$g_1(x_1, \dots, x_{n-1}) \leqslant g_2(x_1, \dots, x_{n-1}) \quad \text{for all } (x_1, \dots, x_{n-1}) \in A'.$$

  In that case, A is called a normal area in $\mathbb{R}^n$ with respect to the variable x_n. The functions g_1 and g_2 are called the lower and upper bounds of the normal area A, respectively. The set A' is called the projection of A onto the first $n-1$ variables, and it is denoted by $\pi(A)$.

Once normal areas have been defined, we can introduce the concept of a parameter-dependent integral, which is a generalization of the concept of integration for functions of a single variable. Given a normal area A in $\mathbb{R}^n$ with respect to the variable x_n, and a continuous function $f : A \to \mathbb{R}$, the parameter-dependent integral of f over A is defined as:

$$\begin{array}{rrcl} F: & \pi(A) & \longrightarrow & \mathbb{R} \\ & (x_1, \dots, x_{n-1}) & \longmapsto & \displaystyle\int_{g_1(x_1,\dots,x_{n-1})}^{g_2(x_1,\dots,x_{n-1})} f(x_1, \dots, x_{n-1}, x_n)\, dx_n \end{array}$$

In other words, the parameter-dependent integral of f over A is a function that assigns to each point in the projection $\pi(A)$ the value of the integral of f with respect to the variable x_n over the interval defined by the lower and upper bounds of A at that point. In the case of a normal area in $\mathbb{R}^2$ with respect to the variable y, the parameter-dependent integral of f over A is given by:

$$F(x) = \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy$$

If g_1 and g_2 are differentiable and f is partially differentiable with respect to x, then F is also differentiable with:

$$F'(x) = \int_{g_1(x)}^{g_2(x)} \frac{\partial f}{\partial x}(x, y)\, dy + f(x, g_2(x)) \cdot g_2'(x) - f(x, g_1(x)) \cdot g_1'(x).$$

In order to integrate a function over a normal area, we can use the following proposition.

Proposición 7.24. *Let A be a normal area in $\mathbb{R}^n$ with respect to the variable x_n, and let $f : A \to \mathbb{R}$ be a continuous function. If the parameter-dependent integral of f over A is integrable over the projection $\pi(A)$, then f is integrable over A, and we have:*

$$\int_A f = \int_{\pi(A)} F.$$

This proposition allows us to compute the integral of a function over a normal area by first computing the parameter-dependent integral of the function over the normal area, and then integrating that parameter-dependent integral over the projection of the normal area. This is a very useful result, as it allows us to compute multiple integrals by iteratively applying single-variable integration. Some important applications of multiple integrals are the following:

- In 2D, the area of a region $R \subset \mathbb{R}^2$ below a continuos function $f : [a, b] \to \mathbb{R}_0^+$ is the following integral:
$$\text{Area} = \int_a^b f(x)\ dx.$$
- The previous well-known example can be easily extended to 3D, where the volume of a region R in $\mathbb{R}^3$ that is below a continuos function $f : A \to \mathbb{R}_0^+$ defined on a normal area A in $\mathbb{R}^2$ is given by:
$$\text{Volume} = \int_A f.$$
- Given a set $M \subset \mathbb{R}^3$ with volume V and homogeneous density, the center of mass of M, (s_1, s_2, s_3), is given by:
$$s_k = \frac{1}{V} \int_M x_k\ dx_1 dx_2 dx_3 \quad \text{for } k = 1, 2, 3.$$

Changing the Coordinate System

Until know, we have just worked with the cartesian coordinate system. However, we can also work with other coordinate systems, such as the polar coordinate system in $\mathbb{R}^2$ or the cylindrical and spherical coordinate systems in $\mathbb{R}^3$. In order to change the coordinate system, some advanced mathematical tools are needed, such as the concept of “difeomorphism”. However, the following intuitive idea can be followed:

- In polar coordinates, $dA = dx\ dy$ is replaced by:
$$dA = r\ dr\ d\theta$$
- In cylindrical coordinates, $dV = dx\ dy\ dz$ is replaced by:
$$dV = r\ dr\ d\theta\ dz$$
- In spherical coordinates, $dV = dx\ dy\ dz$ is replaced by:
$$dV = \rho^2 \sin\phi\ d\rho\ d\phi\ d\theta$$
where ϕ is the angle between the positive z-axis and the line connecting the origin to the point.

8. Aspects of Measurement Data Analysis

In current scientific research, the analysis of measurement data is crucial for drawing meaningful conclusions. More of one measurement, always under the same conditions, is called a *series of measurements*. The measurement result is the mean value of the series of measurements. There are two types of measurements:

- Direct Measurements: The quantity being measured is directly obtained from the measurement process.
- Indirect Measurements: The quantity being measured is calculated from other directly measured quantities using a mathematical relationship.

However, the accuracy of measurements can be affected by various types of errors. These are:

Gross errors These are errors in the strict sense, often resulting from incorrect behavior on the part of the observer or damaged, unusable measuring instruments. Gross errors are always avoidable.

Systematic deviations Also known as systematic errors, these arise from inaccurate measurement methods or faulty measuring instruments, such as calibration or parallax errors. Systematic deviations can be reduced to a negligible extent through care and high-quality instruments. Correction is possible if the deviations can be readily identified theoretically or empirically.

Random deviations Also known as random or statistical errors, these are fluctuations within a series of measurements when the measurement is repeated. These errors are then analyzed using probability theory and statistics.

In the analysis of measurement data, it is essential to consider how the error has been spread. A regression analysis can be used to determine the relationship between the measured quantity and other variables.

8.1. Mathmatical Statistics

When given a measure, the measurement error should also be provided. There are two options for expressing the measurement error:

- Absolute error: This is the difference between the measured value and the expected true value. It is expressed in the same units as the measured quantity.

- Relative error: This is the ratio of the absolute error to the measured value, often expressed as a percentage.

Each of the measurements in a series of measurements is a random variable which follows a certain probability distribution. Some important values of that random variable X are:

- The mean value μ is the expected value of the random variable, calculated as $\mu = E[X] = \langle X \rangle$.
$$\langle X \rangle = \frac{1}{N} \sum_{i=1}^{N} X_i$$
- The variance σ^2 is a measure of the spread of the random variable, calculated as $\sigma^2 = Var[X]$.
$$Var[X] = \frac{1}{N-1} \sum_{i=1}^{N} (X_i - \langle X \rangle)^2$$
It is important to note that the variance is calculated using $N - 1$ in the denominator, which is known as Bessel's correction. This correction is applied to provide an unbiased estimate of the population variance when using a sample of data.
- The standard deviation σ represents the average amount of variability in the data.
$$\sigma = \sqrt{Var[X]}$$
When the number of measurements N increases, the variance and standard deviation tend to be constant.
- The standard error of the mean (SEM) is a measure of how much the sample mean is expected to vary from the true population mean. It is calculated as:
$$SEM = \sigma_m = \frac{\sigma}{\sqrt{N}}$$
When the number of measurements N increases, the standard error of the mean decreases, indicating that the sample mean is likely to be closer to the true population mean. In addition, given to the square root in the denominator, the standard error of the mean decreases at a slower rate than the standard deviation as N increases.
    - In order to improve the accuracy of the measurement by 3 times, the number of measurements must be increased by $3^2 = 9$ times.
    - In order to improve the accuracy of the measurement by 10 times, the number of measurements must be increased by $10^2 = 100$ times.

  As they reader can see, there is a point where increasing the number of measurements does not significantly improve the accuracy of the measurement.

It has empirically been found that the distribution of measurement errors, specially when the number of measurements is large, can be approximated by a normal distribution, also known as a Gaussian distribution. It has the following interesing properties:

- The mean, median, and mode of a normal distribution are all equal.
- The normal distribution is symmetric around its mean, meaning that the left and right sides of the distribution are mirror images of each other.
- The normal distribution is defined by two parameters: the mean μ and the standard deviation σ. The mean determines the center of the distribution, while the standard deviation determines the spread of the distribution.
- The closer a value is to the mean, the higher its probability density. In fact:

$$P[|X-\mu|<\sigma] \approx 68{,}27\,\%$$
$$P[|X-\mu|<2\sigma] \approx 95{,}45\,\%$$
$$P[|X-\mu|<3\sigma] \approx 99{,}73\,\%$$

8.2. Error Propagation

In indirect measurements, the quantity being measured is calculated from other directly measured quantities using a mathematical relationship. Before moving on to quantities that depend on several measured quantities, we will first consider the case of a quantity f that depends on a single measured quantity x. The error in f can be approximated using the total differential of f with respect to x:

$$\Delta f = \left|\frac{df}{dx}\right| \Delta x$$

However, there are cases where the quantity f depends on several measured quantities $x_1, x_2, \ldots, x_n$. In such cases, the error in f just because of the errors in one of the measured quantities x_i can be approximated using the partial derivative of f with respect to x_i:

$$(\Delta f)_i = \left|\frac{\partial f}{\partial x_i}\right| \Delta x_i$$

However, when considering the total error in f due to the errors in all the measured quantities, there are two common approaches:

- The maximum error approach, which simply sums up the individual errors from each measured quantity:

$$(\Delta f)_{\text{máx}} = \sum_{i=1}^{n} (\Delta f)_i = \sum_{i=1}^{n} \left|\frac{\partial f}{\partial x_i}\right| \Delta x_i$$

- Error measurements are often assumed to be independent, and from an statistical point of view, they partially cancel each other out. In this case, the

Gaussian error propagation formula is used, which combines the individual errors in quadrature:

$$(\Delta f)_{\text{Gaussian}} = \sqrt{\sum_{i=1}^{n} (\Delta f)_i^2} = \sqrt{\sum_{i=1}^{n} \left(\left| \frac{\partial f}{\partial x_i} \right|^2 \cdot \Delta x_i^2 \right)}$$

8.3. Regression Analysis

Regression analysis is a statistical method used to model the relationship between a dependent variable and one or more independent variables. Let us name the dependent variable as y and the independent variables as $x \in \mathbb{R}^\ell$, where ℓ is the number of independent variables. Lets say we have a set of N measurements of the dependent variable, denoted as $y_1, y_2, \ldots, y_N$, corresponding to the independent variable values $x_1, x_2, \ldots, x_N$ (taking into account that the independent variable can be a vector). The goal of regression analysis is to find a function $f(x)$ that best fits the data points (x_i, y_i) for $i = 1, 2, \ldots, N$. That function has n parameters, which we will denote as $a_1, a_2, \ldots, a_n$. Then, our model is given by:

$$f : \mathbb{R}^\ell \times \mathbb{R}^n$$

It should be noted that our aim is to find the parameters $a_1, a_2, \ldots, a_n$ so that the function f depends only on the independent variable x.

8.3.1. Linear Regression

Linear regression is a specific type of regression analysis where the model function f is linear in the parameters. This means that f can be expressed as:

$$f(x, a) = \Phi(x) \cdot a = \sum_{j=1}^{n} \Phi_j(x) a_j$$

where $\Phi : \mathbb{R}^\ell \to \mathbb{R}^n$ is a vector of basis functions such thtat $\Phi_1(x) = 1$, and a is the vector of parameters. It should be noted that the basis functions $\Phi_j(x)$ can be nonlinear functions of the independent variable x, but the model is still considered linear because it is linear in the parameters a_j.

Ideally, we would like to find the parameters a such that the model function $f(x_i, a)$ perfectly fits the data points y_i for all i. For each data point, we can stablish the following equation:

$$y_i = f(x_i, a) = \Phi(x_i) \cdot a$$

We will then have a system of N equations (one for each data point) with n unknowns (the parameters a_j). That system can be written in matrix form as:

$$Ka = y$$

where K is the $N \times n$ matrix whose i-th row is given by $\Phi(x_i)$, and y is the vector of measurements. However, given that having more data points improves the accuracy of the measurement, it is common to have $N \gg n$. In this case, it is not possible to find a set of parameters a that perfectly fits all the data points:

$$Ka - y \neq 0 \qquad \forall a \in \mathbb{R}^n$$

Our goal is then to find the parameters a that minimize the error between the model predictions and the actual measurements, i.e., we want to find $a^* \in \mathbb{R}^n$ such that:

$$\|Ka^* - y\| = \min_{a \in \mathbb{R}^n} \|Ka - y\|$$

In order to get rid of the square root, our problem can be equivalently written as finding $a^* \in \mathbb{R}^n$ such that:

$$\|Ka^* - y\|^2 = \min_{a \in \mathbb{R}^n} \|Ka - y\|^2$$

We have just described the *least squares* problem, which is a common method for performing linear regression. Analiticaly, the solution to the least squares problem can be found by finding the minimum of the function:

$$\begin{aligned} F : \quad \mathbb{R}^n &\longrightarrow \mathbb{R} \\ a &\longmapsto \|Ka - y\|^2 = \sum_{i=1}^{N} (f(x_i, a) - y_i)^2 \end{aligned}$$

That can be done by ensuring that $\nabla F(a) = 0$, which has n equations (each of the partial derivatives must be zero) with n unknowns (the parameters a_j).

Observación. In MATLAB, the least squares problem can be solved using the backslash operator, which is a convenient way to solve linear systems of equations. Given a matrix K and a vector y, you can find the parameters a that minimize the least squares error by using `a = K\y`

Deciding which model fits better

When we have several models to choose from, it may be difficult to decide which one fits the data better. One common approach is to use the coefficient of determination, denoted as R^2, which measures the proportion of the variance in the dependent variable that is predictable from the independent variables.

$$R^2 = \frac{\sum_{i=1}^{N} (f(x_i, a) - \langle y \rangle)^2}{\sum_{i=1}^{N} (y_i - \langle y \rangle)^2}$$

where $\langle y \rangle$ is the mean of the observed data y_i. The value of R^2 ranges from 0 to 1, where a value closer to 1 indicates a better fit of the model to the data.

Observación. It should be noted that a high R^2 only informs that the model fits the data *we currently have* well, but it does not necessarily mean that the model will predict future data accurately[1]. Therefore, it is important to consider other factors, such as the complexity of the model and the underlying physical principles, when deciding which model is more appropriate for a given set of data.

[1] "With enough parameters, I can fit an elephant" - John von Neumann

8.3.2. Nonlinear Regression

Nonlinear regression is a type of regression analysis where the model function f is nonlinear in the parameters. This means that f cannot be expressed as a linear combination of the parameters. Nonlinear regression models can take various forms, but the more relevants are the exponential and logarithmic models. These models can be linearized by applying specific transformations to the data, allowing us to use linear regression techniques to estimate the parameters. For example, consider the exponential model:

$$f(x, a) = a_1 e^{a_2 x} a$$

By taking the natural logarithm of both sides, we can linearize the model:

$$\ln(f(x, a)) = \ln(a_1) + a_2 x$$

This transformation allows us to apply linear regression techniques to estimate the parameters $\ln(a_1)$ and a_2. We can then exponentiate the estimated value of $\ln(a_1)$ to obtain the estimate for a_1.

9. Series Expansions: Taylor and Fourier

Definición 9.1 (Sequence)**.** A sequence in a set X is a function $\alpha : \mathbb{N} \to X$. We denote $\alpha(n)$ by x_n and write the sequence as:

$$\{x_n\}_{n\in\mathbb{N}}$$

Definición 9.2 (Series)**.** Given a sequence $\{x_n\}_{n\in\mathbb{N}}$, the series associated to it is the sequence of partial sums:

$$\{s_n\}_{n\in\mathbb{N}} \quad \text{where} \quad s_n = \sum_{k=1}^{n} x_k$$

The series is written as $\sum\limits_{n\geqslant 1} x_n$. It is said to be convergent if the sequence of partial sums $\{s_n\}$ converges, and in that case, we define:

$$\sum_{n=1}^{\infty} x_n = \lim_{n\to\infty} s_n$$

An specific type of series is the **power series**, which has the form:

$$\sum_{n=0}^{\infty} a_n(x-c)^n$$

where $a_n \in \mathbb{R}$ (or $\mathbb{C}$) are the coefficients, c is the center of the series, and x is the variable. The convergence of power series depends on the value of x and can be analyzed using the radius of convergence.

Definición 9.3 (Radius of Convergence)**.** The radius of convergence R of a power series $\sum\limits_{n\geqslant 0} a_n(x-c)^n$ is defined as:

$$R = \frac{1}{\limsup\limits_{n\to\infty} \sqrt[n]{|a_n|}}$$

If the limit superior is zero, we set $R = \infty$, and if it is infinite, we set $R = 0$.

Teorema 9.1. *Let* $\sum\limits_{n\geqslant 0} a_n(x-c)^n$ *be a power series with radius of convergence* R*. Then:*

- *The series converges for all x such that $\|x - c\| < R$.*
- *The series diverges for all x such that $\|x - c\| > R$.*
- *The behavior of the series at the boundary points where $\|x - c\| = R$ must be analyzed separately, as it may converge or diverge depending on the specific coefficients a_n.*

Proposición 9.2. *Given a power series $\sum_{n\geqslant 0} a_n(x-c)^n$, if $a_n \neq 0$ for all $n \in \mathbb{N}$ and the following limit exists, then the radius of convergence can be computed as:*

$$R = \lim_{n\to\infty} \left| \frac{a_n}{a_{n+1}} \right|$$

9.1. Taylor Series

Definición 9.4 (Taylor Series)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be a function that is infinitely differentiable at a point $c \in \mathbb{R}$. The Taylor series of f centered at $c \in \mathbb{R}$ of order n is the power series given by:

$$T_{f,c}(x)^n := \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k$$

where $f^{(k)}(c)$ denotes the k-th derivative of f evaluated at c.

Teorema 9.3 (Taylor's Theorem)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be a function that is $n+1$ times differentiable on an interval containing c and x. Then there exists some ξ between c and x such that:*

$$f(x) = T_{f,c}(x)^n + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-c)^{n+1}$$

In other words, the Taylor series approximates the function f around the point c, and the error of this approximation is given by the remainder term:

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-c)^{n+1}$$

Taylor functions are really used to approximate functions around a specific point, and they are particularly useful when the function is difficult to compute directly.

9.1.1. Approximating a Pendulum as a Spring-Mass System

Consider a simple spring-mass system, where a mass m is attached to a spring with spring constant k (which depends on the stiffness of the spring). The equation of motion for this system is given by Hooke's Law:

$$F_R = -kx$$

where F_R is the restoring force exerted by the spring, and x is the displacement of the mass from its equilibrium position. The negative sign indicates that the force is

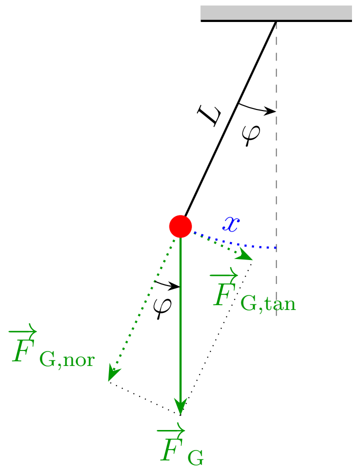

Figura 9.1: Pendulum diagram with forces and angles.

directed opposite to the displacement, trying to restore the mass to its equilibrium position. We will now consider a pendulum, which consists of a mass m attached to a string of length L, swinging under the influence of gravity, as described in Figure 9.1.

The force acting on the mass in the pendulum is the gravitational force, which can be decomposed into two components: one perpendicular to the string (normal component) and one tangential to the arc of motion (tangential component). The tangential component of the gravitational force is responsible for the motion of the pendulum and is given by:

$$F_{\text{G,tan}} = -mg\sin(\varphi)$$

where the negative sign indicates that the force is directed opposite to the direction of motion, and φ is the angle between the string and the vertical. Given that φ is given in radians, we can easily obtain its value:

$$\varphi \cdot L = x \Longrightarrow \varphi = \frac{x}{L}$$

Substituting this into the expression for the tangential force, we get:

$$F_{\text{G,tan}} = -mg\sin\left(\frac{x}{L}\right)$$

Now, we can use the Taylor series expansion of the sine function around 0 (which is the equilibrium position of the pendulum) up to the first order term:

$$\sin\left(\frac{x}{L}\right) \approx \sin\left(\frac{0}{L}\right)(x-0)^0 + \cos\left(\frac{0}{L}\right) \cdot \frac{1}{L}(x-0)^1 = \frac{x}{L}$$

Substituting this approximation back into the expression for the tangential force, we get:

$$F_{\text{G,tan}} \approx -mg \cdot \frac{x}{L} = -\frac{mg}{L}x$$

This shows that for small angles (small displacements), the pendulum behaves like a spring-mass system with an effective spring constant $k_{\text{eff}} = \dfrac{mg}{L}$.

9.2. Fourier Series

Fourier series are used to represent periodic functions as an infinite sum of sines and cosines. They are particularly useful in solving problems in physics and engineering, such as heat conduction, wave propagation, and signal processing.

Definición 9.5 (Periodic Function)**.** A function $f : \mathbb{R} \to \mathbb{R}$ is said to be periodic with period $T > 0$ if for all $x \in \mathbb{R}$, we have:

$$f(x+T) = f(x)$$

Definición 9.6 (Fourier Series)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be a periodic function with period T. The Fourier series of f, denoted by $S_f(x)$, is the infinite series given by:

$$S_f(x) = \frac{a_0}{2} + \sum_{n \geqslant 1} \left[a_n \cos(\omega_n x) + b_n \sin(\omega_n x)\right]$$

where $\omega_n = \frac{2\pi n}{T}$ is the angular frequency, and the coefficients a_n and b_n are given by:

$$\begin{aligned}
a_0 &= \frac{2}{T} \int_0^T f(x)\, dx \\
a_n &= \frac{2}{T} \int_0^T f(x) \cos(\omega_n x)\ dx \quad \text{for } n \geqslant 1 \\
b_n &= \frac{2}{T} \int_0^T f(x) \sin(\omega_n x)\ dx \quad \text{for } n \geqslant 1
\end{aligned}$$

Teorema 9.4. *Let $f : \mathbb{R} \to \mathbb{R}$ be a periodic function with period T. If, within a single period, f satisfies the following conditions:*

1. *f is piecewise continuous and piecewise monotonic (it can be divided into a finite number of subintervals where it is continuous and monotonic).*
2. *f has only jump discontinuities (the left-hand and right-hand limits exist at every point).*

Then, the Fourier series of f converges for all $x \in \mathbb{R}$ to the value:

$$S_f(x) = \frac{f(x^+) + f(x^-)}{2}$$

Where:

- *If f is **continuous** at x, then $S_f(x) = f(x)$.*
- *If x is a **point of discontinuity**, the series converges to the arithmetic mean of the limits from the right and left.*

Proposición 9.5. *Let $f : \mathbb{R} \to \mathbb{R}$ be a periodic function with period T.*

- *If f is an even function, then $b_n = 0$ for all $n \geqslant 1$, and the Fourier series reduces to a cosine series:*

$$S_f(x) = \frac{a_0}{2} + \sum_{n \geqslant 1} a_n \cos(\omega_n x)$$

- *If f is an odd function, then $a_n = 0$ for all $n \geqslant 0$, and the Fourier series reduces to a sine series:*

$$S_f(x) = \sum_{n \geqslant 1} b_n \sin(\omega_n x)$$

9.2.1. Complex Exponential Form of Fourier Series

The Euler's formula, for every $x \in \mathbb{R}$, states that:

$$\begin{aligned} e^{ix} &= \cos(x) + i\sin(x) \\ e^{-ix} &= \cos(x) - i\sin(x) \end{aligned}$$

Using these identities, we can express the sine and cosine functions in terms of complex exponentials:

$$\begin{aligned} \cos(x) &= \frac{e^{ix} + e^{-ix}}{2} = \frac{1}{2} \cdot \left(e^{ix} + e^{-ix}\right) \\ \sin(x) &= \frac{e^{ix} - e^{-ix}}{2i} = -\frac{i}{2} \cdot \left(e^{ix} - e^{-ix}\right) \end{aligned}$$

Substituting these expressions into the Fourier series, we can rewrite it in terms of complex exponentials:

$$S_f(x) = \frac{a_0}{2} + \sum_{n \geqslant 1} \left[\frac{a_n}{2} \cdot \left(e^{i\omega_n x} + e^{-i\omega_n x}\right) - \frac{b_n i}{2} \cdot \left(e^{i\omega_n x} - e^{-i\omega_n x}\right)\right]$$

This can be further simplified defining the complex Fourier coefficients c_n for every $n \in \mathbb{Z}$ as follows:

$$\begin{aligned} c_0 &= \frac{a_0}{2} \\ c_n &= \frac{a_n - ib_n}{2} \\ c_{-n} &= \frac{a_n + ib_n}{2} \end{aligned}$$

With these definitions, the Fourier series can be expressed in its complex exponential form:

$$S_f(x) = \sum_{n \in \mathbb{Z}} c_n e^{i\omega_n x}$$

where $\omega_n = \frac{2\pi n}{T}$ is the angular frequency corresponding to the n-th term of the series and c_n are the complex Fourier coefficients that can be computed using the formula:

$$c_n = \frac{1}{T} \int_0^T f(x) e^{-i\omega_n x}\, dx$$

Working with the complex exponential form of the Fourier series can simplify calculations and provide deeper insights into the properties of the function being represented, especially in the context of signal processing and harmonic analysis.

10. Differential Equations

Differential equations are mathematical equations that relate a function to its derivatives. They are fundamental in describing various physical phenomena, such as motion, heat transfer, and population dynamics. In this chapter, we will explore the basics of differential equations, including their classification, methods of solution, and applications. There are two main types of differential equations:

- Ordinary Differential Equations (ODEs): These involve functions of a single variable and their derivatives.
- Partial Differential Equations (PDEs): These involve functions of multiple variables and their partial derivatives.

Definición 10.1 (Order of a Differential Equation)**.** The order of a differential equation is the highest order of derivative that appears in the equation.

Some common examples of ODEs include:

- The simple harmonic oscillator. According to Hooke's Law, the restoring force F_R exerted by a spring is proportional to the displacement x of the mass from its equilibrium position:
  $$F_R = -kx$$
  where k is the spring constant. According to Newton's second law, the force is also equal to the mass m times the acceleration (the second derivative of displacement with respect to time):
  $$F_R = m\frac{d^2x}{dt^2}$$
  Combining these two equations gives us the second-order ordinary differential equation:
  $$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$
- The Radiative Decay Law. This law ensures that the rate of decay of a radioactive substance is inversely proportional to the amount of substance present. Mathematically, this can be expressed as:
  $$\frac{dN}{dt} = -\lambda N$$
  where $N(t)$ is the number of radioactive atoms at time t, and λ is the decay constant.

- Free fall under gravity. The motion of an object in free fall can be described by the second-order ordinary differential equation:

$$\frac{d^2y}{dt^2} = -g$$

  where $y(t)$ is the height of the object at time t, and g is the acceleration due to gravity.

- When working with capacitors and inductors in electrical circuits, we often encounter first-order ODEs.

It should be noted that a ODE does not usually have a unique solution, as it may contain arbitrary constants that can take on any value. To determine a unique solution, we need to specify initial conditions or boundary conditions.

10.1. ODE solution methods

To solve ordinary differential equations, we can use various methods depending on the type and complexity of the equation.

10.1.1. Primitive finding

The simplest method to solve an ODE is to find its primitive. This works for first-order ODEs of the form:

$$\frac{dy}{dx} = f(x)$$

In this case, we can integrate both sides with respect to x to find the solution:

$$y(x) = \int f(x)\ dx + C$$

where C is the constant of integration.

10.1.2. Separation of variables

For first-order ODEs that can be expressed in the form:

$$\frac{dy}{dx} = g(x)h(y)$$

we can use the method of separation of variables. This involves rearranging the equation to separate the variables:

$$\int \frac{1}{h(y)}\ dy = \int g(x)\ dx$$

This allows us to integrate both sides independently to find the solution. The solution is usually expressed in implicit form, and we may need to solve for y explicitly if possible.

10.2. Types of ODE

Once we have a general understanding of how to solve ODEs, we can classify them into different types based on their structure and properties. We will then explore specific methods for solving each type of ODE.

10.2.1. Linear ODE of first order

A first-order linear ODE has the form:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

In order to solve this type of equation, a distinction is made between the homogeneous case, where $Q(x) = 0$, and the non-homogeneous case, where $Q(x) \neq 0$.

Homogeneous case

In the homogeneous case, the equation simplifies to:

$$\frac{dy}{dx} + P(x)y = 0$$

As we can see, this is a separable equation. Solving it gives us the general solution:

$$y(x) = Ce^{-\int P(x)\,dx}$$

Non-homogeneous case

In the non-homogeneous case, there are two approaches to find the solution:

- If we can find a particular solution $y_p(x)$ to the non-homogeneous equation, then the general solution can be expressed as:

  $$y(x) = y_h(x) + y_p(x)$$

  where $y_h(x)$ is the general solution to the corresponding homogeneous equation. Finding $y_p(x)$ may not be easy, but there are tables that give hints on how can it be.

- Alternatively, once $y_h(x)$ is known, the constant C in the homogeneous solution can be replaced with a function $K(x)$, leading to the method of variation of constant. This involves substituting

  $$y(x) = K(x)e^{-\int P(x)\,dx}$$

  into the original non-homogeneous equation to derive an equation for $K(x)$, which can then be solved to find its value. Once $K(x)$ is determined, the general solution to the non-homogeneous equation is also knwon.

10.2.2. Linear ODE with constant coefficients

A linear ODE with constant coefficients has the form:

$$\frac{d^n y}{dx^n} + a_{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1\frac{dy}{dx} + a_0 y = g(x)$$

where $a_0, a_1, \ldots, a_{n-1}$ are constants and $g(x)$ is a function of x.

Homogeneous case

In the homogeneous case, where $g(x) = 0$, the solutions form a vector space of dimension n. To find the general solution, we can use the characteristic equation:

$$\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0 = 0$$

The roots of this characteristic equation determine the form of the general solution.

- For every distinct (real or complex) root λ_i, there is a corresponding solution of the form:
$$y_i(x) = e^{\lambda_i x}$$
- For every repeated root λ of multiplicity m, there are m linearly independent solutions of the form:
$$y_i(x) = e^{\lambda x}, \quad y_{i+1}(x) = xe^{\lambda x}, \quad \ldots, \quad y_{i+m-1}(x) = x^{m-1}e^{\lambda x}$$
- For every complex solution $z(t)$ of the ODE, two real solutions are known:
$$y_1(t) = \Re(z(t)), \quad y_2(t) = \Im(z(t))$$

That way, we find n different real solutions. In order to check that they are linearly independent, we can compute the Wronskian determinant.

$$W(y_1, y_2, \ldots, y_n)(t) = \begin{vmatrix} y_1(t) & y_2(t) & \cdots & y_n(t) \\ y_1'(t) & y_2'(t) & \cdots & y_n'(t) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(t) & y_2^{(n-1)}(t) & \cdots & y_n^{(n-1)}(t) \end{vmatrix} \qquad t \in I$$

If the Wronskian is non-zero for some t in the interval I, then the solutions are linearly independent. Therefore, the general solution to the homogeneous equation can be expressed as a linear combination of these independent solutions.

Non-homogeneous case

In the non-homogeneous case, where $g(x) \neq 0$, variation of constant could be used to find the general solution. Alternatively, if $g(x)$ is of a specific form (e.g., polynomial, exponential, sine, or cosine), we can use the method of undetermined coefficients to find a particular solution $y_p(x)$. Once we have $y_p(x)$, the general solution to the non-homogeneous equation can be expressed as:

$$y(x) = y_h(x) + y_p(x)$$

11. Übungs Blätter

11.1. Einführung in MATLAB

These exercises are about introducing MATLAB as a tool for numerical computations. They can be found here.

11.2. Grundlagen der Vektor- und Matrizenrechnung

Ejercicio 11.2.1 (Vektoroperationen)**.** Gegeben die Vektoren $\overrightarrow{a} = \begin{pmatrix} 2 \\ 5 \\ 3 \end{pmatrix}$, $\overrightarrow{b} = \begin{pmatrix} -1 \\ 5 \\ 8 \end{pmatrix}$ und $\overrightarrow{c} = \begin{pmatrix} 5 \\ 0 \\ 1 \end{pmatrix}$. Berechnen Sie die Komponenten (Vektorkoordinaten) sowie Betrag und Richtungswinkel des Vektors

$$\overrightarrow{s} = -3\overrightarrow{b} + 4(\overrightarrow{a} \cdot \overrightarrow{c})\overrightarrow{a} - (\overrightarrow{a} - \overrightarrow{b}) \cdot (2\overrightarrow{c} + \overrightarrow{a})\overrightarrow{c}$$

Ejercicio 11.2.2 (Produkte mit Vektoren)**.** Gegeben die Vektoren $\overrightarrow{a} = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$, $\overrightarrow{b} = \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}$ und $\overrightarrow{c} = \begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix}$. Berechnen Sie mit diesen Vektoren die folgenden Produkte:

1. $\overrightarrow{a} \cdot \overrightarrow{b}$
2. $(\overrightarrow{a} - \overrightarrow{c}) \cdot (\overrightarrow{a} + \overrightarrow{c})$
3. $\overrightarrow{b} \times \overrightarrow{c}$
4. $(\overrightarrow{a} \times \overrightarrow{c}) \cdot \overrightarrow{b}$

Ejercicio 11.2.3 (Lineare (Un-)Abhängigkeit)**.** Stellen Sie fest, ob die Vektoren $\overrightarrow{a}$, $\overrightarrow{b}$ und $\overrightarrow{c}$ linear unabhängig oder linear abhängig sind:

1. $\overrightarrow{a} = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$, $\overrightarrow{b} = \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix}$ und $\overrightarrow{c} = \begin{pmatrix} -2 \\ 4 \\ 1 \end{pmatrix}$
2. $\overrightarrow{a} = \begin{pmatrix} 5 \\ 1 \\ 1 \end{pmatrix}$, $\overrightarrow{b} = \begin{pmatrix} -1 \\ 0 \\ 5 \end{pmatrix}$ und $\overrightarrow{c} = \begin{pmatrix} -13 \\ -2 \\ 13 \end{pmatrix}$

Ejercicio 11.2.4 (Orthogonalität von Vektoren)**.** Eine Kraft $\overrightarrow{F} = \begin{pmatrix} F_x \\ F_y \\ F_z \end{pmatrix}$ vom Betrag $F = 100\sqrt{14}$ N soll senkrecht auf einer Ebene E stehen, die durch die Vektoren $\overrightarrow{a} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ und $\overrightarrow{b} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ aufgespannt wird. Bestimmen Sie die (skalaren) Komponenten F_x, F_y und F_z dieser Kraft. Wieviele Lösungen gibt es?

Ejercicio 11.2.5 (Matrizenmultiplikation)**.**

1. Zeigen Sie am Beispiel der folgenden beiden Matrizen

$$A = \begin{pmatrix} -6 & 9 & -3 \\ -4 & 6 & -2 \\ -2 & 3 & -1 \end{pmatrix} \quad \text{und} \quad B = \begin{pmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ 2 & -3 & 0 \end{pmatrix},$$

   dass die Matrizenmultiplikaton nicht-kommutativ ist, d.h. $A \cdot B \neq B \cdot A$.

2. Gegeben die folgenden 3 Matrizen:

$$A_{(3,2)} = \begin{pmatrix} 1 & 3 \\ 2 & 0 \\ 1 & 4 \end{pmatrix}, \qquad B_{(2,2)} = \begin{pmatrix} 2 & 1 \\ 3 & -2 \end{pmatrix} \quad \text{und} \quad C_{(2,3)} = \begin{pmatrix} 2 & 0 & -1 \\ 6 & 1 & 3 \end{pmatrix}.$$

   a) Bilden Sie alle möglichen Produkte $X \cdot Y$ aus zwei Faktoren.

   b) Welche Produkte $X \cdot Y \cdot Z$ aus drei verschiedenen Faktoren können Sie damit bilden?

Ejercicio 11.2.6 (Lösungsmengen linearer Gleichungssysteme)**.** Bestimmen Sie die Lösungsmenge des Gleichungssystems $A\overrightarrow{x} = \overrightarrow{b}$ mit

$$A = \begin{pmatrix} 1 & -1 & 2 & 1 & 0 \\ -2 & 4 & -2 & -1 & 3 \\ 0 & -2 & -2 & -1 & -3 \\ 1 & -2 & 2 & 1 & 1 \end{pmatrix}, \quad \overrightarrow{b} = \begin{pmatrix} 1 \\ 4 \\ -6 \\ 1 \end{pmatrix}.$$

Was können Sie zur Lösbarkeit des Gleichungssystems sagen?

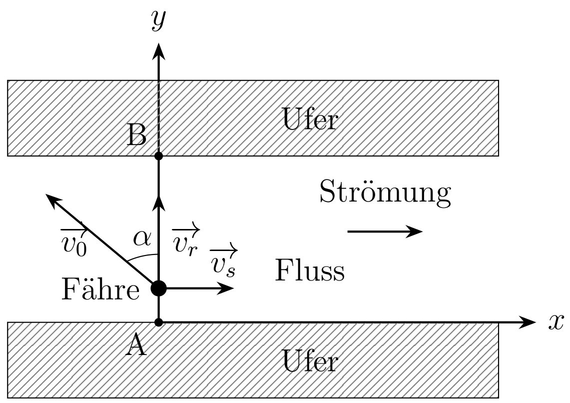

Figura 11.1: Fährverbindung über einen Fluss mit Strömung.

11.3. Rechnen mit physikalischen Größen

Ejercicio 11.3.1 (Fährverbindung)**.** Eine Fähre bewegt sich mit der Eigengeschwindigkeit $v_0 = 4\,{}^{\mathrm{m}}\!/\!{}_{\mathrm{s}}$ (relativ zum Fluss) vom Uferpunkt A aus auf kürzestem Weg zum gegenüberliegenden Flussufer (Punkt B; Abb. 11.1).

1. Unter welchem Winkel α muss die Fähre gegen die Strömung gesteuert werden, wenn die Geschwindigkeit der Strömung den Betrag $v_s = 1{,}5\,{}^{\mathrm{m}}\!/\!{}_{\mathrm{s}}$ hat?

   The Ferry wants to get to B in an straight line. Therefore, it must compensate the flow velocity by steering at an angle α against the flow. It is then needed that:

   $$v_s = v_0 \sin(\alpha) \implies \alpha = \arcsin\left(\frac{v_s}{v_0}\right) = \arcsin\left(\frac{1{,}5}{4}\right) \approx 0{,}3844\,\mathrm{rad} \approx 22{,}024\,^\circ$$

2. Wie groß ist dann die resultierende Geschwindigkeit v_r der Fähre?

   The resulting velocity v_r of the ferry is given by:

   $$v_r = v_0 \cos(\alpha) = 4\cos(0{,}3844) \approx 3{,}7081\,{}^{\mathrm{m}}\!/\!{}_{\mathrm{s}}$$

Ejercicio 11.3.2 (Elektrische Punktladungen im Koordinatensystem)**.** Eine Punktladung $q_1 = 6{,}0\,\mu\mathrm{C}$ befindet sich in einem kartesischen Koordinatensystem bei $x_1 = 1{,}0\,\mathrm{m}$, $y_1 = 0{,}5\,\mathrm{m}$. Eine zweite Ladung $q_2 = -2{,}5\,\mu\mathrm{C}$ befindet sich in dessen Ursprung. Ein Elektron, d.h. eine dritte Punktladung, ist in einem Punkt mit den Koordinaten (x_e, y_e). Berechnen Sie die Werte für x_e und y_e, bei denen sich das Elektron im Gleichgewicht befindet, d.h. bei dem die Gesamtkraft auf das Elektron verschwindet.

Option 1

The forces acting on the electron due to the other two charges must cancel each other out for equilibrium. Let F_{ei} be the force on the electron due to charge q_i. The forces can be expressed as:

$$\vec{F}_{e1} + \vec{F}_{e2} = 0$$

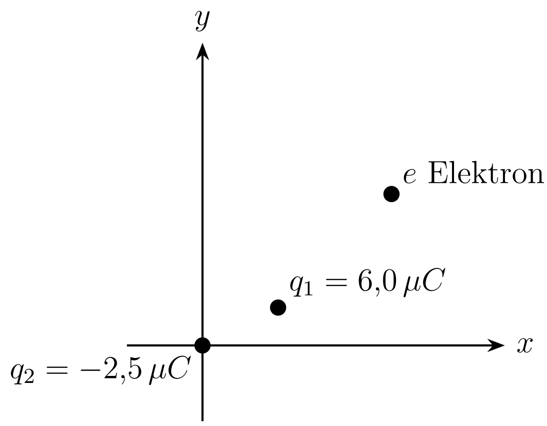

Figura 11.2: Punktladungen im Koordinatensystem.

where, using that $q_e = -e < 0$ (the charge of the electron):

$$\overrightarrow{F}_{e1} = -k_e \frac{|q_1 e|}{r_{e1}^2}\ \hat{r}_{e1}, \quad \overrightarrow{F}_{e2} = k_e \frac{|q_2 e|}{r_{e2}^2}\ \hat{r}_{e2}$$

Here, k_e is Coulomb's constant, e is the elementary charge, r_{ei} is the distance between the electron and charge q_i, and $\hat{r}_{ei}$ is the unit vector pointing from charge q_i to the electron. Using the values of r_{e1} and r_{e2} based on the coordinates of the charges and the electron, we have:

$$\overrightarrow{r}_{e1} = (x_e - 1, y_e - 0{,}5), \quad \overrightarrow{r}_{e2} = (x_e, y_e)$$

Therefore, the equilibrium condition becomes:

$$-k_e \frac{|q_1 e|}{((x_e-1)^2 + (y_e-0{,}5)^2)} \cdot \frac{(x_e - 1, y_e - 0{,}5)}{\sqrt{(x_e-1)^2 + (y_e-0{,}5)^2}} + k_e \frac{|q_2 e|}{(x_e^2 + y_e^2)} \cdot \frac{(x_e, y_e)}{\sqrt{x_e^2 + y_e^2}} = 0$$

This vector equation can be separated into its x and y components, leading to a system of two equations with two unknowns (x_e and y_e). Solving this system will yield the coordinates of the electron in equilibrium.

$$\begin{aligned}
-k_e \frac{|q_1 e|(x_e - 1)}{((x_e-1)^2 + (y_e-0{,}5)^2)^{3/2}} + k_e \frac{|q_2 e| x_e}{(x_e^2 + y_e^2)^{3/2}} &= 0 \\
-k_e \frac{|q_1 e|(y_e - 0{,}5)}{((x_e-1)^2 + (y_e-0{,}5)^2)^{3/2}} + k_e \frac{|q_2 e| y_e}{(x_e^2 + y_e^2)^{3/2}} &= 0
\end{aligned}$$

In an easier way:

$$\begin{aligned}
\frac{|q_1|(x_e - 1)}{((x_e-1)^2 + (y_e-0{,}5)^2)^{3/2}} &= \frac{|q_2| x_e}{(x_e^2 + y_e^2)^{3/2}} \\
\frac{|q_1|(y_e - 0{,}5)}{((x_e-1)^2 + (y_e-0{,}5)^2)^{3/2}} &= \frac{|q_2| y_e}{(x_e^2 + y_e^2)^{3/2}}
\end{aligned}$$

As we can see, this system of equations is nonlinear and may require numerical methods or iterative approaches to solve for x_e and y_e. We should then approach the problem using another method.

Option 2

Given that $\overrightarrow{F}_{e1} = -\overrightarrow{F}_{e2}$, the forces must have the same magnitude but opposite directions. This implies that the electron must lie along the line connecting the two charges, with $x_e, y_e < 0$. Therefore:

$$y_e = \frac{y_1}{x_1} x_e = \frac{1}{2} x_e$$

On the other hand, equalizing the magnitudes of the forces:

$$|\overrightarrow{F}_{e1}| = |\overrightarrow{F}_{e2}| \Longrightarrow k_e \frac{|q_1 e|}{r_{e1}^2} = k_e \frac{|q_2 e|}{r_{e2}^2} \Longrightarrow \frac{|q_1|}{r_{e1}^2} = \frac{|q_2|}{r_{e2}^2} \Longrightarrow \frac{|q_1|}{(x_e-1)^2 + (y_e - 0{,}5)^2} = \frac{|q_2|}{x_e^2 + y_e^2}$$

Therefore, using the relation for y_e in terms of x_e, we have the following equation to solve for x_e:

$$\frac{|q_1|}{(x_e-1)^2 + \left(\frac{1}{2}x_e - 0{,}5\right)^2} = \frac{|q_2|}{x_e^2 + \left(\frac{1}{2}x_e\right)^2}$$

$$6 \cdot \left(\frac{5}{4} x_e^2\right) = 2{,}5 \cdot \left(\frac{5}{4} x_e^2 - 2{,}5 x_e + 1{,}25\right)$$

$$4{,}375 x_e^2 + 6{,}25 x_e - 3{,}125 = 0$$

$$x_e = \frac{-6{,}25 \pm \sqrt{(-6{,}25)^2 - 4 \cdot 4{,}375 \cdot (-3{,}125)}}{2 \cdot 4{,}375} = \frac{-5 \pm 2\sqrt{15}}{7}$$

Given that $x_e < 0$, we take:

$$x_e = -\frac{5 + 2\sqrt{15}}{7} \approx -1{,}8208\,\text{m}$$

$$y_e = \frac{1}{2} x_e = -\frac{5 + 2\sqrt{15}}{14} \approx -0{,}9104\,\text{m}$$

Ejercicio 11.3.3 (Die magnetische Kraft)**.** Ein punktförmiges Teilchen mit einer Ladung $q = -3{,}64\,\text{nC}$ bewegt sich mit einer Geschwindigkeit $\overrightarrow{v} = 2{,}75 \cdot 10^6\,\text{m/s}\ \overrightarrow{e_x}$, d.h. entlang der x-Achse. Berechnen Sie die Kraft, die folgende Felder auf das Teilchen ausüben:

1. $\overrightarrow{B} = 0{,}38\,\text{T}\ \overrightarrow{e_y}$

$$\overrightarrow{F} = q \cdot (\overrightarrow{v} \times \overrightarrow{B}) = q \cdot \begin{vmatrix} \overrightarrow{e_x} & \overrightarrow{e_y} & \overrightarrow{e_z} \\ v_x & 0 & 0 \\ 0 & B_y & 0 \end{vmatrix} = q \cdot v_x \cdot B_y \cdot \overrightarrow{e_z} =$$
$$= -3{,}64 \cdot 10^{-9} \cdot 2{,}75 \cdot 10^6 \cdot 0{,}38\ \overrightarrow{e_z} = -3{,}8038 \cdot 10^{-3}\,\text{N}\ \overrightarrow{e_z}$$

2. $\overrightarrow{B} = T\,0{,}75\ \overrightarrow{e_x} + T\,0{,}75\ \overrightarrow{e_y}$

$$\overrightarrow{F} = q \cdot (\overrightarrow{v} \times \overrightarrow{B}) = q \cdot \begin{vmatrix} \overrightarrow{e_x} & \overrightarrow{e_y} & \overrightarrow{e_z} \\ v_x & 0 & 0 \\ B_x & B_y & 0 \end{vmatrix} = q \cdot v_x \cdot B_y \cdot \overrightarrow{e_z} =$$
$$= -3{,}64 \cdot 10^{-9} \cdot 2{,}75 \cdot 10^6 \cdot 0{,}75\ \overrightarrow{e_z} = -7{,}5075 \cdot 10^{-3}\,\text{N}\ \overrightarrow{e_z}$$

11.4. Komplexe Zahlen

Ejercicio 11.4.1 (Eine Rechenaufgabe mit komplexen Zahlen)**.** Gegeben seien drei komplexe Zahlen $z_1 = 2 - i$, $z_2 = 5 + 2i$ und $z_3 = -3i$. Berechne:

$$\frac{\overline{z_1} + 2z_3 \cdot z_1 - iz_2}{2(z_2 - \overline{z_3}) \cdot z_1}$$

Ejercicio 11.4.2 (Rechnen mit komplexen Zahlen)**.**

1. Berechnen Sie Realteil, Imaginärteil und Betrag der komplexen Zahl

$$z = 3 \cdot \overline{1 + 2i} \cdot (5 + i) + 5 \cdot \frac{28i - 3}{14 + (1 + 2i)^3}.$$

2. Bestimmen Sie alle $z \in \mathbb{C}$, die folgende Gleichung lösen:

$$z \cdot \overline{z} + (3 - 4i)z + (3 + 4i)\overline{z} + 9 = 0.$$

   Beschreiben Sie die Gestalt der Lösungsmenge in der Gaußschen Zahlenebene.

3. Berechnen Sie Realteil, Imaginärteil, Argument und Betrag von $z \in \mathbb{C}$:

$$z = 2^{-30} \cdot \left(i - \sqrt{3}\right)^{32}.$$

4. Bestimmen Sie Betrag und Argument aller $v \in \mathbb{C}$, die im 2. Quadranten liegen (d.h. $\Re(z) \leqslant 0$, $\Im(z) \geqslant 0$) und folgende Gleichung lösen:

$$v^6 = -1 + i.$$

5. Berechnen Sie

$$z = z_1 + z_2 + z_3 = 2 \cdot e^{-5/4i\pi} + 0{,}2(1 - 3i)^3 + \frac{2 - i}{3 + 4i}.$$

   Geben Sie das Ergebnis sowohl in kartesischer als auch in exponenzieller Form an.

Ejercicio 11.4.3 (Beträge komplexer Zahlen)**.** Beweisen oder widerlegen Sie: Für alle komplexen Zahlen gilt:

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2 \cdot \left(|z_1|^2 + |z_2|^2\right).$$

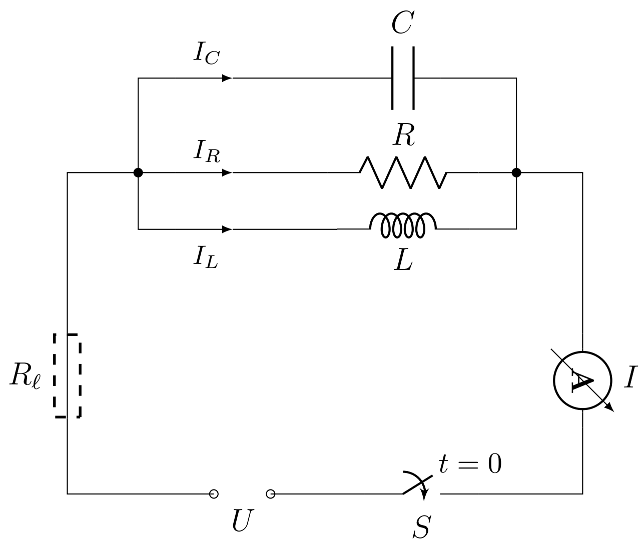

Figura 11.3: Skizze zur Aufgabe 11.5.2

11.5. Anwendung komplexer Rechnung

Ejercicio 11.5.1. Zeigen Sie durch Rechnung in komplexer Form: Drei gleichfrequente, sinusförmige Wechselströme i_1, i_2 und i_3 mit den Amplituden i_0, der Kreisfrequenz $\omega > 0$ und den Nullphasenwinkeln $\varphi_1 = 0$, $\varphi_2 = {}^2\!/\!_3\pi$ und $\varphi_3 = {}^4\!/\!_3\pi$ löschen sich bei ungestörter Überlagerung gegenseitig aus, d.h. $i_1 + i_2 + i_3 = 0$.

Let $\underline{i}_1(t)$, $\underline{i}_2(t)$ und $\underline{i}_3(t)$ be:

$$\begin{aligned}
\underline{i}_1(t) &= i_0 \cdot e^{\jmath(\omega t+0)} = i_0 \cdot e^{\jmath\omega t} \\
\underline{i}_2(t) &= i_0 \cdot e^{\jmath(\omega t+2/3\pi)} = i_0 \cdot e^{\jmath\omega t} \cdot e^{\jmath 2/3\pi} \\
\underline{i}_3(t) &= i_0 \cdot e^{\jmath(\omega t+4/3\pi)} = i_0 \cdot e^{\jmath\omega t} \cdot e^{\jmath 4/3\pi}
\end{aligned}$$

Therefore:

$$\begin{aligned}
\underline{i}_1(t) + \underline{i}_2(t) + \underline{i}_3(t) &= i_0 \cdot e^{\jmath\omega t}\left(e^{\jmath 0} + e^{\jmath 2/3\pi} + e^{\jmath 4/3\pi}\right) = \\
&= i_0 \cdot e^{\jmath\omega t}\left(1 + \left(-\frac{1}{2} + \jmath\frac{\sqrt{3}}{2}\right) + \left(-\frac{1}{2} - \jmath\frac{\sqrt{3}}{2}\right)\right) = \\
&= i_0 \cdot e^{\jmath\omega t} \cdot 0 = 0
\end{aligned}$$

Therefore, we have shown that:

$$i_1(t) + i_2(t) + i_3(t) = \Im\left(\underline{i}_1(t) + \underline{i}_2(t) + \underline{i}_3(t)\right) = \Im(0) = 0$$

Ejercicio 11.5.2 (Resonanz im Parallelschwingkreis)**.** Der in Abbildung 11.3 skizzierte Parallelschwingkreis mit dem ohmschen Widerstand $R = 10\,\Omega$, der Induktivität $L =\ 0{,}2\,\mathrm{H}$ und der Kapazität $C = 10\,\mathrm{mF}$ wird durch eine Wechselstromquelle mit dem Effektivwert $I = 10\,\mathrm{A}$ und der variablen Kreisfrequenz ω zu elektromagnetischen Schwingungen angeregt.

1. Im Resonanzfall sind der Gesamtstrom I und die angelegte Spannung U in Phase, d.h. $\varphi_I - \varphi_U = 0$. Bei welcher Kreisfrequenz ω_0 tritt dieser Fall ein? Wie groß ist dann der komplexe Gesamtwiderstand Z?

   The complex resistance is:

$$\underline{Z} = \frac{\underline{u}(t)}{\underline{i}(t)} = \frac{\sqrt{2}Ue^{\jmath(\omega t+\varphi_U)}}{\sqrt{2}Ie^{\jmath(\omega t+\varphi_I)}} = \frac{U}{I}e^{\jmath(\varphi_U-\varphi_I)} \overset{(*)}{=} \frac{U}{I}e^{\jmath 0} = \frac{U}{I} \in \mathbb{R}$$

   where in $(*)$ we used the fact that in resonance $\varphi_U = \varphi_I$.

   We can calculate the complex resistance as:

$$\begin{aligned}\frac{1}{\underline{Z}} &= \frac{1}{\underline{Z}_R} + \frac{1}{\underline{Z}_L} + \frac{1}{\underline{Z}_C} \\ &= \frac{1}{R} + \frac{1}{\jmath\omega L} + \jmath\omega C \\ &\Longrightarrow \underline{Z} = \frac{1}{\frac{1}{R} - \jmath\frac{1}{\omega L} + \jmath\omega C} = \frac{1}{\frac{1}{R} + \jmath\left(-\frac{1}{\omega L} + \omega C\right)}\end{aligned}$$

   Given that $\underline{Z} \in \mathbb{R}$, we have that the imaginary part must be zero:

$$\begin{aligned}-\frac{1}{\omega_0 L} + \omega_0 C = 0 &\Longrightarrow \omega_0 C = \frac{1}{\omega_0 L} \Longrightarrow \omega_0^2 = \frac{1}{LC} \Longrightarrow \\ &\Longrightarrow \omega_0 = \sqrt{\frac{1}{LC}} = \sqrt{\frac{1}{0{,}2 \cdot 10 \cdot 10^{-3}}} = \sqrt{500} \approx 22{,}36\,{}^{\mathrm{rad}}\!/\!{}_{\mathrm{s}}\end{aligned}$$

   Therefore, we have that:

$$\underline{Z} = Z = R = 10\,\Omega$$

2. Wie ändert sich die Situation, wenn die Zuleitungen nicht mehr als ideal angenommen werden und einen ohmschen Widerstand R_ℓ beitragen? Welche Werte haben ω_0 und Z dann?

   In this case, we have:

$$\underline{Z} = \frac{1}{\frac{1}{R} + \jmath\left(-\frac{1}{\omega L} + \omega C\right)} + R_\ell$$

   Therefore, ω_0 does not change, but:

$$\underline{Z} = Z = R + R_\ell = 10 + R_\ell$$

Ejercicio 11.5.3 (Wechselstrommessbrücke)**.** Mit der in Abbildung 11.4 dargestellten Brückenschaltung lässt sich ein unbekannter komplexer Widerstand $\underline{Z}_1 = \underline{Z}_X$ wie folgt bestimmen: Bei vorgegebenen (komplexen) Widerständen $\underline{Z}_2$ und $\underline{Z}_3$ wird der stetig veränderbare komplexe Widerstand $\underline{Z}_4$ so eingestellt, dass der Brückenzweig A–B stromlos wird. Das in die Brücke geschaltete Wechselstromamperemeter mit dem (bekannten) Innenwiderstand $\underline{Z}_5$ dient dabei lediglich als Nullindikator.

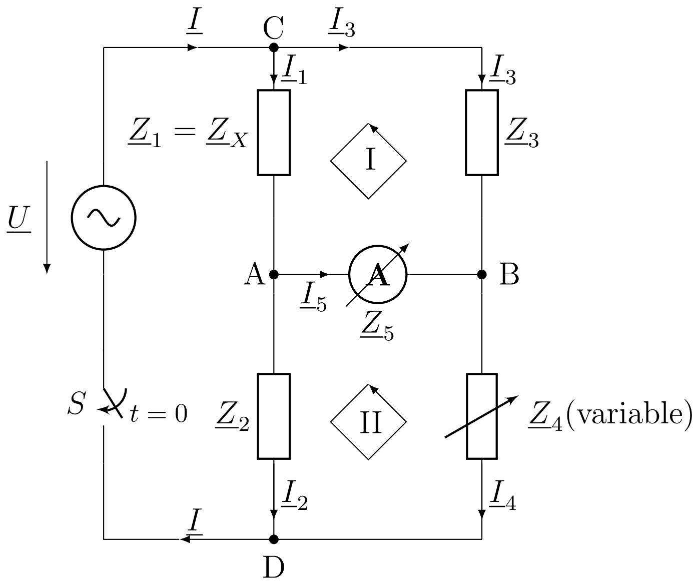

Figura 11.4: Skizze zur Aufgabe 11.5.3.

1. Wie lautet die sogenannte Abgleichbedingung, d.h. die Bedingung für die Stromlosigkeit des Brückenzweiges A-B?

   Let's denote $\underline{U}_{ij} := \underline{U}_i - \underline{U}_j$ the voltage between nodes i and j. According to Kirchhoff's laws, we have that:

$$\underline{U}_{CA} + \underline{U}_{AB} - \underline{U}_{CB} = 0$$
$$\underline{U}_{AD} - \underline{U}_{BD} - \underline{U}_{AB} = 0$$

   In order that the branch A-B is currentless, we need:

$$0 = \underline{I}_5 = \frac{\underline{U}_{AB}}{\underline{Z}_5} \iff \underline{U}_{AB} = 0$$

   Therefore, we have:

$$\underline{U}_{CA} = \underline{U}_{CB} \qquad \underline{U}_{AD} = \underline{U}_{BD} \tag{11.1}$$

   Using now Kirchhoff's Intensity Law on nodes A and B, taking into account that $\underline{I}_5 = 0$, we have:

$$\underline{I}_1 = \underline{I}_2$$
$$\underline{I}_3 = \underline{I}_4$$

   Using Ohm's Law, we have:

$$\frac{\underline{U}_{CA}}{\underline{Z}_1} = \frac{\underline{U}_{AD}}{\underline{Z}_2} \qquad \frac{\underline{U}_{CB}}{\underline{Z}_3} = \frac{\underline{U}_{BD}}{\underline{Z}_4} \tag{11.2}$$

   Unifying Equations (11.1) and (11.2), we obtain:

$$\frac{\underline{U}_{CA}}{\underline{Z}_1} = \frac{\underline{U}_{AD}}{\underline{Z}_2} \Longrightarrow \frac{\underline{U}_{CA}}{\underline{U}_{AD}} = \frac{\underline{Z}_1}{\underline{Z}_2}$$
$$\frac{\underline{U}_{CA}}{\underline{Z}_3} = \frac{\underline{U}_{AD}}{\underline{Z}_4} \Longrightarrow \frac{\underline{U}_{CA}}{\underline{U}_{AD}} = \frac{\underline{Z}_3}{\underline{Z}_4}$$

We should denote that we can divide by $\underline{U}_{AD}$ because if $\underline{U}_{AD} = 0$, then $\underline{U}_{BD} = \underline{U}_{AD} = 0$ and therefore $\underline{I}_2 = \underline{I}_4 = 0$, which according to the Kirchhoff's Intensity Law implies that $\underline{I} = 0$, which would mean that $\underline{U} = 0$, which is not the case.

Therefore, we have the balance condition:

$$\frac{\underline{Z}_1}{\underline{Z}_2} = \frac{\underline{Z}_3}{\underline{Z}_4} \Longrightarrow \underline{Z}_1\underline{Z}_4 = \underline{Z}_2\underline{Z}_3$$

2. In einem konkreten Fall haben die festen Widerstände $\underline{Z}_2$ und $\underline{Z}_3$ folgende Werte:

$$\underline{Z}_2 = 10\Omega - \jmath \cdot 2\Omega,$$
$$\underline{Z}_3 = 8\Omega - \jmath \cdot 6\Omega.$$

Die Brücke A–B wird dabei genau dann stromlos, wenn der variable Widerstand $\underline{Z}_4$ auf den Wert

$$\underline{Z}_4 = 5\Omega - \jmath \cdot 2\Omega$$

eingestellt wird. Welchen Wert besitzt dann der (zunächst noch unbekannte) Widerstand $\underline{Z}_X$?

Using the balance condition obtained in the previous part, we obtain:

$$\begin{aligned} \underline{Z}_X = \underline{Z}_1 &= \frac{\underline{Z}_2\underline{Z}_3}{\underline{Z}_4} \\ &= \frac{(10 - 2\jmath)(8 - 6\jmath)}{5 - 2\jmath} = \\ &= \frac{492}{29}\Omega - \jmath \cdot \frac{244}{29}\Omega \approx 16{,}97\Omega - \jmath \cdot 8{,}41\Omega \end{aligned}$$

Ejercicio 11.5.4 (Wechselstromparadoxon)**.** Der in Bild 11.5 dargestellte Wechselstromkreis enthält die ohmschen Widerstände R und R_x und einen zu R_x parallel geschalteten Kondensator mit der Kapazität C. Beim Anlegen einer Wechselspannung $\underline{U}$ mit der Kreisfrequenz ω fließt der Gesamtstrom $\underline{I}$, dessen Effektivwert I durch das zugeschaltete Wechselstrommessgerät A gemessen wird. Der Innenwiderstand R_i des Geräts sei im Widertstand R bereits enthalten. Zeigen Sie: Der ohmsche Widerstand R_x lässt sich so wählen, dass die Stromanzeige unabhängig ist von der Stellung des Schalters S (geschlossen oder offen; sog Wechselstromparadoxon).

We have two options:

- Switch open: In this case, everything is connected in serial and $\underline{I}_1 = 0$. Therefore:

$$\underline{Z}_{\text{open}} = R + \frac{1}{\jmath\omega C} = R - \frac{1}{\omega C} \cdot \jmath$$

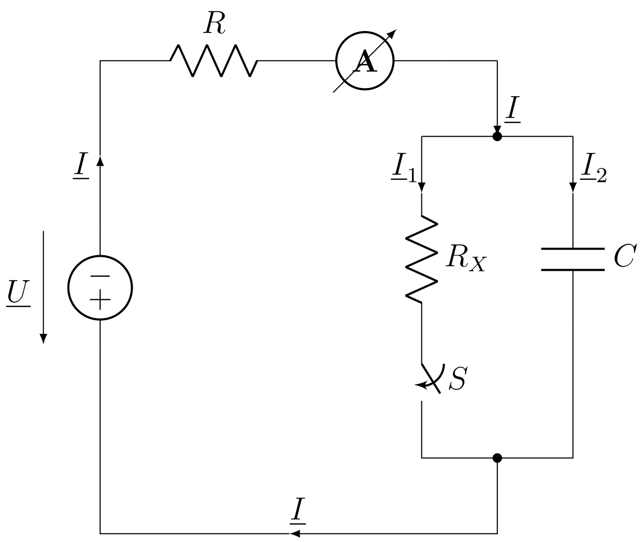

Figura 11.5: Skizze zur Aufgabe 11.5.4.

- Switch closed:

$$\begin{aligned}
\underline{Z}_{\text{closed}} &= R + \frac{1}{\frac{1}{R_X} + \jmath\omega C} = \left(R + \frac{\frac{1}{R_X}}{\frac{1}{R_X^2} + \omega^2 C^2}\right) - \frac{\omega C}{\frac{1}{R_X^2} + \omega^2 C^2} \cdot \jmath = \\
&= \left(R + \frac{R_X}{1 + (R_X\omega C)^2}\right) - \frac{R_X^2 \omega C}{1 + (R_X\omega C)^2} \cdot \jmath = \\
&= \frac{R(1 + (R_X\omega C)^2) + R_X}{1 + (R_X\omega C)^2} - \frac{R_X^2\omega C}{1 + (R_X\omega C)^2} \cdot \jmath
\end{aligned}$$

In order that the current does not depend on the position of the switch, it is needed:

$$\widehat{I}_{\text{closed}} = \widehat{I}_{\text{open}} \Longrightarrow |\underline{I}_{\text{closed}}| = |\underline{I}_{\text{open}}| \Longrightarrow \left|\frac{\underline{U}_{\text{closed}}}{\underline{Z}_{\text{closed}}}\right| = \left|\frac{\underline{U}_{\text{open}}}{\underline{Z}_{\text{open}}}\right|$$

Given that $\underline{U}_{\text{open}} = \underline{U}_{\text{closed}}$, we only need that:

$$\begin{aligned}
&|\underline{Z}_{\text{closed}}| = |\underline{Z}_{\text{open}}| \Longrightarrow |\underline{Z}_{\text{closed}}|^2 = |\underline{Z}_{\text{open}}|^2 \Longrightarrow \\
\Longrightarrow & R^2 + \frac{1}{\omega^2 C^2} = \frac{(R(1 + (R_X\omega C)^2) + R_X)^2 + (R_X^2\omega C)^2}{(1 + (R_X\omega C)^2)^2} \Longrightarrow \\
\Longrightarrow & (R(1 + (R_X\omega C)^2))^2 + \frac{(1 + (R_X\omega C)^2)^2}{\omega^2 C^2} = \\
& \qquad = (R(1 + (R_X\omega C)^2))^2 + R_X^2 + 2R_X\ R(1 + (R_X\omega C)^2) + (R_X^2\omega C)^2 \Longrightarrow \\
\Longrightarrow & \frac{(1 + (R_X\omega C)^2)^2}{\omega^2 C^2} = R_X^2 + 2R_X\ R(1 + (R_X\omega C)^2) + (R_X^2\omega C)^2 \Longrightarrow \\
\Longrightarrow & 1 + 2(R_X\omega C)^2 = (R_X\omega C)^2 + 2\omega^2 C^2 R_X\ R(1 + (R_X\omega C)^2) \Longrightarrow \\
\Longrightarrow & 1 + (R_X\omega C)^2 = 2\omega^2 C^2 R_X\ R(1 + (R_X\omega C)^2) \Longrightarrow \\
\Longrightarrow & (2\omega^2 C^2 R_X\ R - 1)(1 + (R_X\omega C)^2) = 0
\end{aligned}$$

Therefore, the only possible solution is:

$$R_X = \frac{1}{2R(\omega C)^2}$$

Using that value of R_X, we will measure the same current value with the switch opened and closed.

11.6. Differenzialrechnung von Funktionen einer Veränderlichen

Ejercicio 11.6.1 (Ableitung trigonometrischer Funktionen mit Produktregel)**.** Differenzieren Sie die folgenden Funktionen nach der Variablen x:

1. $f(x) = 3x^2 \cdot \cos(x)$
2. $f(x) = \sin(x) \cdot \cos(x)$

Ejercicio 11.6.2 (Quotientenregel)**.** Differenzieren Sie die folgenden Funktionen nach der Variablen x und bestimmen Sie den größtmöglichen Definitionsbereich in $\mathbb{R}$:

1. $f(x) = 2 \cdot \frac{\ln(x)}{x}$
2. $f(x) = 2 \cdot \frac{\cos(x)}{3x^2}$

Ejercicio 11.6.3 (Kettenregel)**.** Differenzieren Sie nach der Variablen x:

1. $f(x) = 3(2x+1)^4$
2. $f(x) = e^x$
3. $f(x) = e^{2x}$
4. $f(x) = e^{2x^3}$

Ejercicio 11.6.4 (Gemischte Aufgaben zur Differenzialrechnung)**.** Differenzieren Sie nach der Variablen x und geben Sie ggf. Einschränkungen des Definitionsbereichs an, falls die Differenzierbarkeit nicht auf ganz $\mathbb{R}$ gegeben ist.

1. $f(x) = x^{x^2}$
2. $f(x) = (x+1)^x$
3. $f(x) = x^{(x+1)}$
4. $f(x) = (x+1)^{(x+1)}$
5. $f(x) = \left(2x^2+1\right)^{\sin(x)}$
6. $f(x) = \frac{e^{-\cos x}}{\sin(x)}$

Ejercicio 11.6.5 (Differenzieren in 1D - Test)**.** Differenzieren Sie nach der Variablen x. Welche Bedingungen muss der jeweilige Definitionsbereich erfüllen?

1. $f(x) = \left(2x^2 - \frac{4}{x} + 3\right)^4$
2. $f(x) = e^{2x\cos x}$
3. $f(x) = \ln\left(\frac{1}{x^2}\right)$
4. $f(x) = \sqrt{\frac{x^2-a^2}{a^2+x^2}}$

Ejercicio 11.6.6 (Kurvendiskussion)**.** Diskutieren Sie den Verlauf der Funktion

$$f(x) = \frac{(x-2)^2}{x+2}.$$

Machen Sie Aussagen zu Definitionsbereich, Symmetrie, Nullstellen, Polen, Ableitungen (bis zur 3. Ordnung), relative Extremwerte, Wende- und Sattelpunkte und dem Verhalten "im Unendlichen". Fertigen Sie am Schluss eine saubere Skizze des Kurvenverlaufs an.

11.7. Integralrechnung von Funktionen einer Veränderlichen

Ejercicio 11.7.1 (Partielle Integration oder Substitution?)**.** Berechnen Sie die folgenden unbestimmten Integrale.

1. $\displaystyle\int (x+5)e^x\,dx$
2. $\displaystyle\int e^{\cos x}\sin x\,dx$

Ejercicio 11.7.2 (Zwei bestimmte Integrale)**.** Berechnen Sie die folgenden Integrale mit partieller Integration oder Substitution.

1. $\displaystyle\int_0^1 x^2\cdot\sqrt{2x+1}\,dx$
2. $\displaystyle\int_0^\pi \cos^3 x\cdot\sin x\,dx$

Ejercicio 11.7.3 (Logarithmische Integration)**.** Berechnen Sie:

1. $\displaystyle\int \frac{dx}{x\cdot\ln x}$
2. $\displaystyle\int \frac{3x^2-2}{2x^3-4x+2}\,dx$

Ejercicio 11.7.4 (Fläche zwischen Sinus und Kosinus)**.** Berechnen Sie den Inhalt der Fläche, die zwischen zwei benachbarten Schnittpunkten der Sinuskurve und der Kosinuskurve von den beiden Kurven eingeschlossen wird.

Ejercicio 11.7.5 (Integral einer gebrochen-rationalen Funktion)**.** Berechnen Sie:

$$\int \frac{x+3}{x^2-1}\,dx$$

Ejercicio 11.7.6 (Partialbruchzerlegung)**.** Berechnen Sie eine Stammfunktion für:

$$f(x)=\frac{5x^3+6x^2-2x+23}{(x+1)^2(x^2-4x+8)}\qquad (x\neq -1)$$

Ejercicio 11.7.7 (Partialbruchzerlegung mit Polynomdivision)**.** Berechnen Sie eine Stammfunktion für:

$$f(x)=\frac{x^5-3x^4+6x^3-3x^2-2x+10}{(x^2-2x+2)^2}$$

11.8. Ableitungen von Funktionen mehrerer Veränderlicher

Ejercicio 11.8.1 (Partielle Ableitungen 1. und 2. Ordnung)**.** Bilden Sie alle partiellen Ableitungen erster und zweiter Ordnung der Funktion:

$$\begin{array}{rrcl} f: & \mathbb{R}^2 & \longrightarrow & \mathbb{R} \\ & (x,y) & \longmapsto & x^2 \cdot e^{-xy} \end{array}$$

The partial derivatives of first order are:

$$\begin{aligned} \frac{\partial f}{\partial x}(x,y) &= 2xe^{-xy} - x^2ye^{-xy} = e^{-xy}(2x - x^2y) \\ \frac{\partial f}{\partial y}(x,y) &= -x^3e^{-xy} \end{aligned}$$

The partial derivatives of second order are:

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2}(x,y) &= \frac{\partial}{\partial x}\left(e^{-xy}(2x - x^2y)\right) = -ye^{-xy}(2x - x^2y) + e^{-xy}(2 - 2xy) \\ \frac{\partial^2 f}{\partial y^2}(x,y) &= \frac{\partial}{\partial y}\left(-x^3e^{-xy}\right) = x^4e^{-xy} \\ \frac{\partial^2 f}{\partial y \partial x}(x,y) &= \frac{\partial}{\partial x}\left(-x^3e^{-xy}\right) = -3x^2e^{-xy} + x^3ye^{-xy} \end{aligned}$$

Ejercicio 11.8.2 (Eine Differenzialgleichung)**.** Zeigen Sie, dass die Funktion

$$\begin{array}{rrcl} f: & \mathbb{R}^2 & \longrightarrow & \mathbb{R} \\ & (x,t) & \longmapsto & e^{-\pi^2a^2t} \cdot \sin(\pi x) \end{array}$$

eine Lösung der folgenden Gleichung ist:

$$a^2\frac{\partial^2 f}{\partial x^2} = \frac{\partial f}{\partial t} \qquad (a \in \mathbb{R} \text{ const.})$$

Let's compute the partial derivatives:

$$\begin{aligned} \frac{\partial f}{\partial t}(x,t) &= -\pi^2a^2e^{-\pi^2a^2t}\sin(\pi x) \\ \frac{\partial^2 f}{\partial x^2}(x,t) &= -\pi^2e^{-\pi^2a^2t}\sin(\pi x) \end{aligned}$$

Therefore, it is clear that it is a solution of the given equation.

Ejercicio 11.8.3 (Totales Differenzial einer gebrochen rationalen Funktion)**.** Bestimmen Sie das totale Differenzial der Funktion

$$\begin{array}{rrcl} f: & \mathbb{R}^2 & \longrightarrow & \mathbb{R} \\ & (x,y) & \longmapsto & \dfrac{x^2 + y^2}{y - x} \end{array}$$

To find the total differential of the function, we first compute the partial derivatives:

$$\begin{aligned}\frac{\partial f}{\partial x}(x,y) &= \frac{2x(y-x)+(x^2+y^2)}{(y-x)^2} = \frac{-x^2+2xy+y^2}{(y-x)^2}\\ \frac{\partial f}{\partial y}(x,y) &= \frac{2y(y-x)-(x^2+y^2)}{(y-x)^2} = \frac{y^2-2xy-x^2}{(y-x)^2}\end{aligned}$$

Therefore the total differential is:

$$\begin{aligned}df(x,y) &= \frac{-x^2+2xy+y^2}{(y-x)^2}dx + \frac{y^2-2xy-x^2}{(y-x)^2}dy =\\ &= \frac{(-x^2+2xy+y^2)\,dx+(y^2-2xy-x^2)\,dy}{(y-x)^2}\end{aligned}$$

11.9. Kurvenintegrale

Ejercicio 11.9.1 (Kurvenintegral erster Art)**.** Berechnen Sie das Kurvenintegral erster Art $\displaystyle\int_{\varphi_1} f\,dt$ für die skalarwertige Funktion

$$\begin{array}{rrcl} f: & \mathbb{R}^3 & \longrightarrow & \mathbb{R} \\ & (x,y,z) & \longmapsto & \sqrt{x^2+y^2+z^2} \end{array}$$

über die parametrisierte Kurve

$$\begin{array}{rrcl} \varphi_1: & [0,2\pi] & \longrightarrow & \mathbb{R}^3 \\ & t & \longmapsto & (t\cos(t), t\sin(t), t) \end{array}$$

First of all, we compute the derivative of the parameterization:

$$\varphi_1'(t) = \begin{pmatrix} \cos(t) - t\sin(t) \\ \sin(t) + t\cos(t) \\ 1 \end{pmatrix}$$

We then realize that φ is differentiable, and given that the third component is constant and non-zero, we can conclude that the parameterization is regular. Therefore, in order to compute the norm of the derivative, we have:

$$\begin{aligned}
\|\varphi_1'(t)\| &= \sqrt{(\cos(t) - t\sin(t))^2 + (\sin(t) + t\cos(t))^2 + 1^2} \\
&= \sqrt{\cos^2(t) - \cancel{2t\sin(t)\cos(t)} + t^2\sin^2(t) + \sin^2(t) + \cancel{2t\sin(t)\cos(t)} + t^2\cos^2(t) + 1} \\
&= \sqrt{1 + t^2 + 1} = \sqrt{t^2+2}
\end{aligned}$$

Therefore, we can now compute the curve integral:

$$\begin{aligned}
\int_{\varphi_1} f\,dt &= \int_0^{2\pi} f(\varphi_1(t))\|\varphi_1'(t)\|\,dt \\
&= \int_0^{2\pi} \sqrt{t^2+t^2}\cdot\sqrt{t^2+2}\,dt \\
&= \sqrt{2}\int_0^{2\pi} t\sqrt{t^2+2}\,dt
\end{aligned}$$

We use the substitution given by:

$$\begin{array}{rrcl} \Phi: & [2, 4\pi^2+2] & \longrightarrow & [0,2\pi] \\ & u & \longmapsto & \sqrt{u-2} \end{array}$$

Thus, using $t = \Phi(u)$, we have:

$$\begin{aligned}
\int_{\varphi_1} f\,dt &= \sqrt{2}\int_{\Phi^{-1}(0)}^{\Phi^{-1}(2\pi)} \Phi(u)\sqrt{\Phi(u)^2+2}\cdot\Phi'(u)\,du \\
&= \sqrt{2}\int_2^{4\pi^2+2} \sqrt{u-2}\sqrt{u}\cdot\frac{1}{2\sqrt{u-2}}\,du = \frac{\sqrt{2}}{2}\int_2^{4\pi^2+2}\sqrt{u}\,du = \frac{\sqrt{2}}{2}\left[\frac{2}{3}u^{3/2}\right]_2^{4\pi^2+2} = \\
&= \frac{\sqrt{2}}{3}\left[u^{3/2}\right]_2^{4\pi^2+2} = \frac{\sqrt{2}}{3}\left((4\pi^2+2)^{3/2} - 2^{3/2}\right)
\end{aligned}$$

Ejercicio 11.9.2 (Kurvenintegral zweiter Art)**.** Berechnen Sie das Kurvenintegral zweiter Art $\displaystyle\int_{\varphi_2} \overrightarrow{f}\, d\overrightarrow{s}$ für das Vektorfeld

$$\begin{array}{rrcl} \overrightarrow{f}: & \mathbb{R}^3 & \longrightarrow & \mathbb{R}^3 \\ & (x,y,z) & \longmapsto & (x-z, y^2, 1) \end{array}$$

über die parametrisierte Kurve

$$\begin{array}{rrcl} \varphi_2: & [0,1] & \longrightarrow & \mathbb{R}^3 \\ & t & \longmapsto & \left(t^2, t, e^t\right) \end{array}$$

We first compute the derivative of the parameterization:

$$\varphi_2'(t) = (2t, 1, e^t)$$

We then realize that ϕ is differentiable, and given that the third component is non-zero, we can conclude that the parameterization is regular. Then, the dot product $\overrightarrow{f}(\varphi_2(t)) \cdot \varphi_2'(t)$ is given by:

$$\begin{aligned} \overrightarrow{f}(\varphi_2(t)) \cdot \varphi_2'(t) &= (t^2 - e^t, t^2, 1) \cdot (2t, 1, e^t) \\ &= 2t^3 - 2te^t + t^2 + e^t \end{aligned}$$

Therefore, we can now compute the curve integral:

$$\begin{aligned} \int_{\varphi_2} \overrightarrow{f}\, d\overrightarrow{s} &= \int_0^1 \overrightarrow{f}(\varphi_2(t)) \cdot \varphi_2'(t)\, dt \\ &= \int_0^1 2t^3 - 2te^t + t^2 + e^t\, dt \end{aligned}$$

Let's firstly compute the integral of te^t using integration by parts:

$$\left[\begin{array}{ll} u(t) = t & u'(t) = 1 \\ v(t) = e^t & v'(t) = e^t \end{array}\right]$$

$$\int_0^1 te^t\, dt = \left[te^t\right]_0^1 - \int_0^1 e^t\, dt$$

Now, we can compute the entire integral:

$$\begin{aligned} \int_{\varphi_2} \overrightarrow{f}\, d\overrightarrow{s} = \left[2 \cdot \frac{t^4}{4} - 2\left(te^t - e^t\right) + \frac{t^3}{3} + e^t\right]_0^1 = \left[\frac{t^4}{2} - 2te^t + 3e^t + \frac{t^3}{3}\right]_0^1 = \\ = \frac{1}{2} - 2e + 3e + \frac{1}{3} - 3 = -\frac{2}{3} + e \end{aligned}$$

Ejercicio 11.9.3 (Länge einer Kurve)**.** Berechnen Sie die Länge der Kurve C mit der Parameterdarstellung

$$\begin{array}{rrcl} \varphi: & [-1,1] & \longrightarrow & \mathbb{R}^2 \\ & t & \longmapsto & (x(t), y(t)) \end{array}$$

$$\begin{array}{rrcl} x: & [-1,1] & \longrightarrow & \mathbb{R} \\ & t & \longmapsto & t^3 \end{array}$$

$$\begin{array}{rrcl} y: & [-1,1] & \longrightarrow & \mathbb{R} \\ & t & \longmapsto & 1-t^2 \end{array}$$

Observación. Man denke an die Voraussetzung zur Längenberechnung mittels Kurvenintegralen, dass die Parameterdarstellung regulär sein muss, d.h. $\varphi'(t) \neq \overrightarrow{0}$. Ist dies nicht für alle Punkte der Kurve der Fall, berechnet man die Länge abschnittsweise.

Given that $x(t)$ is inyective, we realize that φ is also inyective. Let's compute the derivative of the parameterization:

$$\varphi'(t) = (3t^2, -2t)$$

We realize that φ is differentiable, and that $\varphi'(t) = \overrightarrow{0}$ only at $t = 0$. Therefore, we will compute the length of the curve in two parts: in $[-1, 0]$ and in $[0, 1]$. Thus, we compute the norm of the derivative:

$$\|\varphi'(t)\| = \sqrt{(3t^2)^2 + (-2t)^2} = \sqrt{9t^4 + 4t^2} = \sqrt{t^2(9t^2 + 4)} = |t|\sqrt{9t^2 + 4}$$

Therefore, we can now compute the length of the curve:

$$\begin{aligned} L &= \int_{-1}^{0} \|\varphi'(t)\|\, dt + \int_{0}^{1} \|\varphi'(t)\|\, dt \\ &= \int_{-1}^{0} -t\sqrt{9t^2 + 4}\, dt + \int_{0}^{1} t\sqrt{9t^2 + 4}\, dt \end{aligned}$$

We use the substitution given by:

$$\begin{array}{rccc} \Phi : & \mathbb{R} & \longrightarrow & \mathbb{R} \\ & t & \longmapsto & u = 9t^2 + 4 \end{array}$$

In both intervals, Φ es biyective. Thus, using $t = \Phi^{-1}(u)$, we have:

$$\begin{aligned} L &= \int_{\Phi(-1)}^{\Phi(0)} -\Phi^{-1}(u)\sqrt{u} \cdot (\Phi^{-1})'(u)\, du + \int_{\Phi(0)}^{\Phi(1)} \Phi^{-1}(u)\sqrt{u} \cdot (\Phi^{-1})'(u)\, du \\ &= \int_{13}^{4} -\Phi^{-1}(u)\sqrt{u} \cdot \frac{1}{2\Phi^{-1}(u)} \cdot \frac{1}{9}\, du + \int_{4}^{13} \Phi^{-1}(u)\sqrt{u} \cdot \frac{1}{2\Phi^{-1}(u)} \cdot \frac{1}{9}\, du = \\ &= 2 \cdot \frac{1}{18} \int_{4}^{13} \sqrt{u}\, du = \frac{1}{9} \left[\frac{2}{3} u^{3/2}\right]_{4}^{13} = \frac{2}{27} \left(13^{3/2} - 4^{3/2}\right) = \frac{26\sqrt{13} - 16}{27} \end{aligned}$$

Ejercicio 11.9.4 (Flächeninhalt einer Ellipse)**.** Zeigen Sie, dass der Flächeninhalt F einer Ellipse E, die durch die Gleichung

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

beschrieben wird, $F = \pi ab$ ist.

Observación. Verwenden Sie die Parameterdarstellung

$$\begin{array}{rccc} \varphi : & [0, 2\pi] & \longrightarrow & \mathbb{R}^2 \\ & t & \longmapsto & (a\cos t, b\sin t) \end{array}$$

We first compute the derivative of the parameterization:

$$\varphi'(t) = (-a \sin t, b \cos t)$$

We then realize that ϕ is differentiable, and given that $a, b > 0$ and that both $\sin t$ and $\cos t$ cannot be zero at the same time, we can conclude that the parameterization is regular. Given that the curve goes around the ellipse counter-clockwise, we can use the following formula to compute the area:

$$\begin{aligned} F &= \frac{1}{2} \int_{\varphi} (-y, x) \cdot d\overrightarrow{x} = \frac{1}{2} \int_0^{2\pi} (-b \sin t, a \cos t) \cdot (-a \sin t, b \cos t)\, dt \\ &= \frac{1}{2} \int_0^{2\pi} ab \sin^2 t + ab \cos^2 t\, dt = \frac{ab}{2} \int_0^{2\pi} 1\, dt = \frac{ab}{2} \cdot 2\pi = \pi ab \end{aligned}$$

11.10. Potenzialfelder und Mehrfachintegrale

Ejercicio 11.10.1 (Kurvenintegral einer Funktion mit Potenzial)**.** Im $\mathbb{R}^2$ ist eine Kurve gegeben durch die Parameterdarstellung

$$\begin{array}{rrcl} \varphi: & [0, \pi/2] & \longrightarrow & \mathbb{R}^2 \\ & t & \longmapsto & (\sin t, t) \end{array}$$

Berechnen Sie das Kurvenintegral $\displaystyle\int_\varphi \overrightarrow{f} \cdot d\overrightarrow{s}$ für:

$$\begin{array}{rrcl} \overrightarrow{f}: & \mathbb{R}^2 & \longrightarrow & \mathbb{R}^2 \\ & (x,y) & \longmapsto & \begin{pmatrix} 2xy + 2\cos(y) \\ x^2 - 2x\sin(y) + 1 \end{pmatrix} \end{array}$$

Observación. Prüfen Sie, ob $\overrightarrow{f}$ ein Potenzial hat und nutzen Sie es ggf. aus.

Let's first check that φ is a regular parametrization:

$$\varphi'(t) = (\cos t, 1)$$

Given that $1 \neq 0$ for all $t \in [0, \pi/2]$, we conclude that φ is regular. Let's now try to calculate the integral directly. In order to do so, we need to compute $\overrightarrow{f}(\varphi(t)) \cdot \varphi'(t)$:

$$\begin{aligned} \overrightarrow{f}(\varphi(t)) = \overrightarrow{f}(\sin t, t) &= \begin{pmatrix} 2\sin t \cdot t + 2\cos t \\ \sin^2 t - 2\sin^2 t + 1 \end{pmatrix} = \begin{pmatrix} 2t\sin t + 2\cos t \\ 1 - \sin^2 t \end{pmatrix} = \begin{pmatrix} 2t\sin t + 2\cos t \\ \cos^2 t \end{pmatrix} \\ \varphi'(t) &= (\cos t, 1) \\ \overrightarrow{f}(\varphi(t)) \cdot \varphi'(t) &= 2t\sin t\cos t + 2\cos^2 t + \cos^2 t = 2t\sin t\cos t + 3\cos^2 t \end{aligned}$$

Therefore, the integral is:

$$\int_\varphi \overrightarrow{f} \cdot d\overrightarrow{s} = \int_0^{\pi/2} \left(2t\sin t\cos t + 3\cos^2 t\right) dt$$

Given that integrating this expression is quite cumbersome, let's check if $\overrightarrow{f}$ has a potential function. Given that $\mathbb{R}^2$ is star-shaped, we check:

$$\begin{aligned} \frac{\partial f_1}{\partial y} &= 2x - 2\sin(y) \\ \frac{\partial f_2}{\partial x} &= 2x - 2\sin(y) \end{aligned}$$

Since the mixed partial derivatives are equal, $\overrightarrow{f}$ has a potential function. Let's find it:

Option 1 Having fixed $(x,y) \in \mathbb{R}^2$, let's consider the following path from $(0,0)$ to (x,y):

$$\begin{array}{rrcl} \Phi: & [0,1] & \longrightarrow & \mathbb{R}^2 \\ & t & \longmapsto & (tx, ty) \end{array}$$

Let's also write the potential function as $u : \mathbb{R}^2 \to \mathbb{R}$. Then:

$$\int_\Phi \overrightarrow{f} \cdot d\overrightarrow{s} = u(\Phi(1)) - u(\Phi(0)) = u(x,y) \Longrightarrow u(x,y) = \int_0^1 \overrightarrow{f}(\Phi(t)) \cdot \Phi'(t)dt + u(0,0)$$

Given that the potential is defined up to a constant, we can set $u(0,0) = 0$.
Now, let's compute $\overrightarrow{f}(\Phi(t)) \cdot \Phi'(t)$:

$$\begin{aligned}
\overrightarrow{f}(\Phi(t)) &= \overrightarrow{f}(tx,ty) = \begin{pmatrix} 2(tx)(ty) + 2\cos(ty) \\ (tx)^2 - 2(tx)\sin(ty) + 1 \end{pmatrix} = \begin{pmatrix} 2t^2xy + 2\cos(ty) \\ t^2x^2 - 2tx\sin(ty) + 1 \end{pmatrix} \\
\Phi'(t) &= (x,y) \\
\overrightarrow{f}(\Phi(t)) \cdot \Phi'(t) &= 2t^2x^2y + 2x\cos(ty) + t^2x^2y - 2txy\sin(ty) + y \\
&= 3t^2x^2y + 2x\cos(ty) - 2txy\sin(ty) + y
\end{aligned}$$

Therefore, the potential function is:

$$\begin{aligned}
u(x,y) &= \int_0^1 \left(3t^2x^2y + 2x\cos(ty) - 2txy\sin(ty) + y\right) dt = \\
&= \left[x^2yt^3 + \frac{2x}{y}\sin(ty) + yt\right]_0^1 - 2xy\int_0^1 t\sin(ty)dt = \\
&= x^2y + \frac{2x}{y}\sin(y) + y - 2xy\int_0^1 t\sin(ty)dt
\end{aligned}$$

To compute the remaining integral, we use integration by parts:

$$\begin{bmatrix} u(t) = t & u'(t) = 1 \\ v(t) = -\frac{1}{y}\cos(ty) & v'(t) = \sin(ty) \end{bmatrix}$$

$$\begin{aligned}
\int_0^1 t\sin(ty)dt &= \left[-\frac{t}{y}\cos(ty)\right]_0^1 + \frac{1}{y}\int_0^1 \cos(ty)dt = \\
&= \left[-\frac{t}{y}\cos(ty) + \frac{1}{y^2}\sin(ty)\right]_0^1 = -\frac{1}{y}\cos(y) + \frac{1}{y^2}\sin(y)
\end{aligned}$$

Therefore, the potential function is:

$$\begin{aligned}
u(x,y) &= x^2y + \frac{2x}{y}\sin(y) + y - 2xy\left(-\frac{1}{y}\cos(y) + \frac{1}{y^2}\sin(y)\right) = \\
&= x^2y + \cancel{\frac{2x}{y}\sin(y)} + y + 2x\cos(y) - \cancel{\frac{2x}{y}\sin(y)} = \\
&= x^2y + y + 2x\cos(y)
\end{aligned}$$

Option 2 We look for a function $u(x,y)$ such that:

$$\begin{aligned}
\frac{\partial u}{\partial x} &= 2xy + 2\cos(y) \\
\frac{\partial u}{\partial y} &= x^2 - 2x\sin(y) + 1
\end{aligned}$$

Integrating the first equation with respect to x:

$$u(x, y) = \int 2xy + 2\cos(y)\, dx = x^2 y + 2x\cos(y) + h(y)$$

where $h(y)$ is an arbitrary function of y. Now, we differentiate $u(x, y)$ with respect to y and equate it to the second equation:

$$\frac{\partial u}{\partial y} = x^2 - 2x\sin(y) + h'(y) = x^2 - 2x\sin(y) + 1 \Longrightarrow h'(y) = 1 \Longrightarrow h(y) = y + C$$

Given that the potential is defined up to a constant, we can set $C = 0$. Therefore, the potential function is:

$$u(x, y) = x^2 y + 2x\cos(y) + y$$

Finally, we can compute the integral using the potential function:

$$\int_\varphi \overrightarrow{f} \cdot d\overrightarrow{s} = u(\varphi(\pi/2)) - u(\varphi(0)) = u(1, \pi/2) - u(0, 0) = \frac{\pi}{2} + 2\cos(\pi/2) + \frac{\pi}{2} - 0 = \pi$$

Ejercicio 11.10.2 (Doppelintegral mit e-Funktion)**.** Berechnen Sie

$$I = \int_0^1 \int_0^{y^2} e^{x/y}\, dx\, dy$$

$$\int_0^1 \int_0^{y^2} e^{x/y}\, dx\, dy = \int_0^1 \left[ye^{x/y}\right]_0^{y^2} dy = \int_0^1 y\left(e^y - 1\right) dy = \int_0^1 ye^y dy - \int_0^1 y dy$$

To compute the remaining integral, we use integration by parts:

$$\begin{bmatrix} u(y) = y & u'(y) = 1 \\ v(y) = e^y & v'(y) = e^y \end{bmatrix}$$

$$\int_0^1 ye^y dy = [ye^y]_0^1 - \int_0^1 e^y dy = [ye^y - e^y]_0^1 = (e - e) - (0 - 1) = 1$$

Therefore, the value of the integral is:

$$I = 1 - \left[\frac{y^2}{2}\right]_0^1 = 1 - \frac{1}{2} = \frac{1}{2}$$

Ejercicio 11.10.3 (Flächenberechnung mit Doppelintegral)**.** Ein Flächenstück wird durch die Kurven $x = 0$, $y = 2x$ $y = 1/a x^2 + a$, $a > 0$ berandet. Berechnen Sie den Flächeninhalt A mit Hilfe eines Doppelintegrals.

In Figure 11.6, we can see the area to be calculated. Let's first find the points of intersection between the curves $y = 2x$ and $y = 1/a x^2 + a$:

$$2x = \frac{1}{a}x^2 + a \Longrightarrow x^2 - 2ax + a^2 = 0 \Longrightarrow (x - a)^2 = 0 \Longrightarrow x = a$$

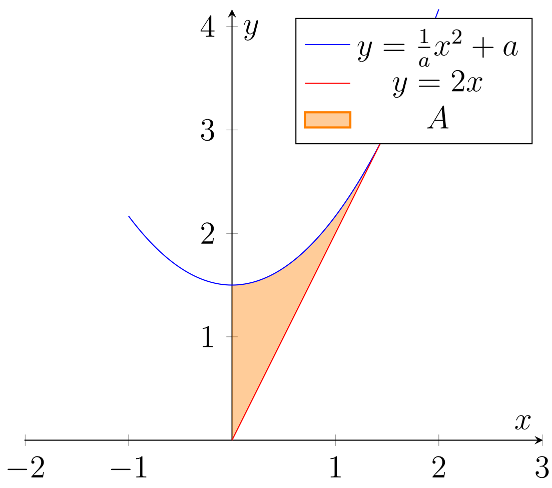

Figura 11.6: Area to be calculated in Exercise 11.10.3.

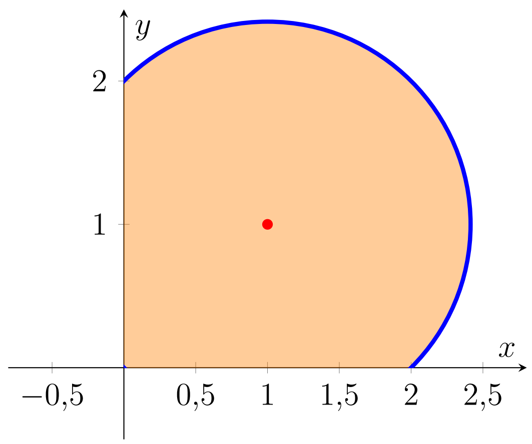

Figura 11.7: Area to be calculated in Exercise 11.10.4.

Therefore, the curves intersect at the point $(a, 2a)$. Given that the area is bounded by $x = 0$ and $x = a$, we can set up the double integral to calculate the area:

$$A = \int_0^a \int_{2x}^{1/ax^2+a} dy\,dx = \int_0^a \left(\frac{1}{a}x^2 + a - 2x\right) dx = \left[\frac{1}{3a}x^3 + ax - x^2\right]_0^a =$$
$$= \frac{1}{3a}a^3 + a^2 - a^2 - 0 = \frac{a^2}{3}$$

Ejercicio 11.10.4 (Flächeninhalt mit Polarkoordinaten)**.** Die Randkurve eines Gebiets in der Ebene wird durch die Gleichung

$$r = 2(\cos(\varphi) + \sin(\varphi))$$

in Polarkoordinaten beschrieben. Berechnen Sie den Flächeninhalt A dieses Gebiets.

In Figure 11.7, we can see the area to be calculated. Using polar coordinates, the area can be calculated as:

$$A = \int_0^{\pi/2} \int_0^{2(\cos(\varphi)+\sin(\varphi))} r\, dr\, d\varphi = \int_0^{\pi/2} \left[\frac{r^2}{2}\right]_0^{2(\cos(\varphi)+\sin(\varphi))} d\varphi = \int_0^{\pi/2} 2(\cos(\varphi) + \sin(\varphi))^2 d\varphi =$$

$$= 2 \int_0^{\pi/2} \left(\cos^2(\varphi) + 2\sin(\varphi)\cos(\varphi) + \sin^2(\varphi)\right) d\varphi = 2 \int_0^{\pi/2} (1 + \sin(2\varphi))\, d\varphi =$$

$$= 2 \left[\varphi - \frac{1}{2}\cos(2\varphi)\right]_0^{\pi/2} = 2\left(\frac{\pi}{2} - \frac{1}{2}\cos(\pi) - \left(0 - \frac{1}{2}\cos(0)\right)\right) = 2\left(\frac{\pi}{2} + \frac{1}{2} + \frac{1}{2}\right) = \pi + 2$$

Ejercicio 11.10.5 (Dreifachintegral in Zylinderkoordinaten)**.** Berechnen Sie das folgende in Zylinderkoordinaten gegebene Integral:

$$I = \int_\pi^{2\pi} \int_0^1 \int_r^{r^2} rz \cdot \sin(\varphi)\, dz\, dr\, d\varphi.$$

$$I = \int_\pi^{2\pi} \int_0^1 \int_r^{r^2} rz\sin(\varphi)\, dz\, dr\, d\varphi = \int_\pi^{2\pi} \int_0^1 \left[\frac{rz^2}{2}\sin(\varphi)\right]_{z=r}^{z=r^2} dr\, d\varphi =$$

$$= \int_\pi^{2\pi} \sin(\varphi) \int_0^1 r \cdot \left[\frac{z^2}{2}\right]_r^{r^2} dr\, d\varphi = \int_\pi^{2\pi} \sin(\varphi) \int_0^1 r \cdot \left(\frac{r^4}{2} - \frac{r^2}{2}\right) dr\, d\varphi =$$

$$= \frac{1}{2} \int_\pi^{2\pi} \sin(\varphi) \int_0^1 \left(r^5 - r^3\right) dr\, d\varphi = \frac{1}{2} \int_\pi^{2\pi} \sin(\varphi) \left[\frac{r^6}{6} - \frac{r^4}{4}\right]_0^1 d\varphi =$$

$$= \frac{1}{2} \int_\pi^{2\pi} \sin(\varphi) \left(\frac{1}{6} - \frac{1}{4}\right) d\varphi = -\frac{1}{24} \int_\pi^{2\pi} \sin(\varphi) d\varphi = \frac{1}{24} [\cos(\varphi)]_\pi^{2\pi} = \frac{1}{24}(1 - (-1)) = \frac{1}{12}$$

11.11. Datenanalyse und Fehlerrechnung

Ejercicio 11.11.1 (Ein neues Schwimmbecken). Bei einem Schwimmbadbauer wird ein quaderförmiges Becken mit einer Länge $l = 50\,\mathrm{m}$, einer Breite $b = 15\,\mathrm{m}$ und einer Tiefe $h = 2\,\mathrm{m}$ in Auftrag gegeben. Erfahrungsgemäß baut die Firma ziemlich ungenau mit statistisch verteilten Fehlern von $\sigma_l = 0{,}4\,\mathrm{m}$, $\sigma_b = 0{,}2\,\mathrm{m}$ sowie $\sigma_h = 0{,}1\,\mathrm{m}$ für Länge, Breite bzw. Tiefe. Berechnen Sie die das Volumen des Schwimmbeckens und den zu erwartenden relativen Fehler nach der Gaußschen Fehlerfortpflanzung. Wieviel Wasser (in m^3) sollte bereitstehen, damit das Becken auf jeden Fall bis zum Rand gefüllt werden kann?

The volume V of the swimming pool is given by the formula:

$$V = l \cdot b \cdot h = 50\,\mathrm{m} \cdot 15\,\mathrm{m} \cdot 2\,\mathrm{m} = 1500\,\mathrm{m}^3$$

To calculate the relative error in the volume using Gaussian error propagation, we use the formula:

$$\begin{aligned}
\sigma_V &= \sqrt{\left(\frac{\partial V}{\partial l}\sigma_l\right)^2 + \left(\frac{\partial V}{\partial b}\sigma_b\right)^2 + \left(\frac{\partial V}{\partial h}\sigma_h\right)^2} = \\
&= \sqrt{(b \cdot h \cdot \sigma_l)^2 + (l \cdot h \cdot \sigma_b)^2 + (l \cdot b \cdot \sigma_h)^2} = \\
&= \sqrt{(15\,\mathrm{m} \cdot 2\,\mathrm{m} \cdot 0{,}4\,\mathrm{m})^2 + (50\,\mathrm{m} \cdot 2\,\mathrm{m} \cdot 0{,}2\,\mathrm{m})^2 + (50\,\mathrm{m} \cdot 15\,\mathrm{m} \cdot 0{,}1\,\mathrm{m})^2} = \\
&= \sqrt{(12\,\mathrm{m}^3)^2 + (20\,\mathrm{m}^3)^2 + (75\,\mathrm{m}^3)^2} = \\
&= \sqrt{144\,\mathrm{m}^6 + 400\,\mathrm{m}^6 + 5625\,\mathrm{m}^6} = \\
&= \sqrt{6169\,\mathrm{m}^6} \approx 78{,}55\,\mathrm{m}^3
\end{aligned}$$

The relative error in the volume is then given by:

$$\frac{\sigma_V}{V} = \frac{78{,}55\,\mathrm{m}^3}{1500\,\mathrm{m}^3} \approx 0{,}0524 \approx 5{,}24\,\%$$

Therefore, the volume of the swimming pool is $1500\,\mathrm{m}^3$ with an expected relative error of approximately $5{,}24\,\%$. To ensure that the pool can be filled to the brim with a level of confidence of around $99{,}7\,\%$ (which corresponds to 3σ), we should prepare the following amount of water:

$$V + 3\sigma_V = 1500\,\mathrm{m}^3 + 3 \cdot 78{,}55\,\mathrm{m}^3 \approx 1735{,}65\,\mathrm{m}^3$$

Ejercicio 11.11.2 (Selbstinduktion einer Doppelleitung). Eine elektrische Doppelleitung besteht aus zwei parallelen, elektrisch leitenden Drähten mit der Länge l und dem Radius r. Der Abstand ihrer Mittelpunkte beträgt a. Die Selbstinduktivität L dieser Doppelleitung in Luft wird dabei nach der Formel

$$L(l, r, a) = \frac{\mu_0}{\pi}\, l \cdot \ln\left(\frac{a-r}{r} + \frac{1}{4}\right)$$

berechnet, mit der magnetischen Feldkonstante $\mu_0 = 4\pi \cdot 10^{-7}\,\mathrm{Vs}/\mathrm{Am}$.

1. Welche Selbstinduktivität L besitzt eine Doppelleitung, deren Dimensionen wie folgt gemessen wurden?

$$l = (2000 \pm 10\,\mathrm{m}); \quad r = (2 \pm 0{,}05\,\mathrm{mm}); \quad a = (30 \pm 0{,}3\,\mathrm{cm})$$

The self-inductance L of the twin-lead can be calculated using the given formula:

$$\begin{aligned} L &= \frac{\mu_0}{\pi}\, l \cdot \ln\left(\frac{a-r}{r} + \frac{1}{4}\right) = \\ &= \frac{4\pi \cdot 10^{-7}\,\mathrm{Vs}/\mathrm{Am}}{\pi} \cdot 2000\,\mathrm{m} \cdot \ln\left(\frac{0{,}30\,\mathrm{m} - 0{,}002\,\mathrm{m}}{0{,}002\,\mathrm{m}} + \frac{1}{4}\right) = \\ &= 8 \cdot 10^{-4}\,\mathrm{Vs}/\mathrm{A} \ln(149 + 0{,}25) \approx 8 \cdot 10^{-4}\,\mathrm{Vs}/\mathrm{A} \cdot 5{,}005622 = \\ &= 4{,}004498\,\mathrm{mH} \end{aligned}$$

2. Wie groß ist die absolute bzw. prozentuale Messunsicherheit des Mittelwertes von L (d. h. der absolute bzw. prozentuale mittlere Fehler des Mittelwertes von L)?

The absolute uncertainty in L can be calculated using Gaussian error propagation:

$$\begin{aligned} \Delta L &= \sqrt{\left(\frac{\partial L}{\partial l}\Delta l\right)^2 + \left(\frac{\partial L}{\partial r}\Delta r\right)^2 + \left(\frac{\partial L}{\partial a}\Delta a\right)^2} = \\ &= \sqrt{\left(\frac{\mu_0}{\pi}\ln\left(\frac{a-r}{r} + \frac{1}{4}\right)\Delta l\right)^2 + \left(\frac{\mu_0}{\pi} l \cdot \frac{1}{\frac{a}{r} - \frac{3}{4}} \cdot \frac{a}{r^2}\Delta r\right)^2 + \left(\frac{\mu_0}{\pi} l \cdot \frac{1}{\frac{a}{r} - \frac{3}{4}} \cdot \frac{1}{r}\Delta a\right)^2} = \\ &= \sqrt{\left(\frac{\mu_0}{\pi}\ln\left(\frac{a}{r} - \frac{3}{4}\right)\Delta l\right)^2 + \left(\frac{\mu_0}{\pi} l \cdot \frac{a}{ar - \frac{3r^2}{4}}\Delta r\right)^2 + \left(\frac{\mu_0}{\pi} l \cdot \frac{1}{a - \frac{3r}{4}}\Delta a\right)^2} \approx \\ &\approx 2{,}95 \cdot 10^{-5}\,\mathrm{H} = \\ &= 0{,}0295\,\mathrm{mH} \end{aligned}$$

The percentage uncertainty in L is then given by:

$$\frac{\Delta L}{L} = \frac{0{,}0295\,\mathrm{mH}}{4{,}004498\,\mathrm{mH}} \approx 0{,}00737 \approx 0{,}737\,\%$$

Ejercicio 11.11.3 (Regressionsanalyse)**.** Gegeben die folgende Wertetabelle:

t	-2	-1	0	1	2
y	13,1	-2	$-2{,}2$	-4	$-15{,}9$

Finden Sie x_1 und x_2 für die beste Kurve der Form $y = f(t) = x_1 + x_2 t^3$ zur Anpassung der Messdaten. Welchen Wert hat das zugehörige Bestimmtheitsmaß R^2?

11.12. Brook Taylor und Joseph Fourier

Ejercicio 11.12.1 (Konvergenzradius einer Potenzreihe)**.** Berechnen Sie den Konvergenzradius r der Potenzreihe

$$s = \sum_{n=2}^{\infty} \frac{x^n}{\sqrt{n^2 - 1}}$$

Ejercicio 11.12.2 (Taylor-Entwicklung mit Wurzelfunktion)**.** Berechnen Sie die Taylorsche Reihe der Funktion

$$f(x) = \frac{1}{\sqrt{x}}$$

am Entwicklungspunkt $x_0 = 1$ bis zur 4. Ordnung und bestimmen Sie den Konvergenzbereich.

Ejercicio 11.12.3 (Relativistische Massenzunahme)**.** Die Masse m eines Elektrons nimmt nach der Relativitätstheorie mit der Geschwindigkeit v zu. Es gilt:

$$m = m(v) = \frac{m_0}{\sqrt{1 - \left(\frac{v}{c}\right)^2}}$$

wobei m_0 die Ruhemasse des Elektrons und c die Lichtgeschwindigkeit ist. Entwickeln Sie mit Hilfe der Potenzreihenentwicklung eine Näherungsformel für die Abhängigkeit zwischen Masse und Geschwindigkeit unter der Annahme $v \ll c$.

Ejercicio 11.12.4 (Freier Fall mit Reibung)**.** Unter Berücksichtigung des Luftwiderstandes besteht zwischen der Fallgeschwindigkeit v und dem Fallweg s der folgende (komplizierte) Zusammenhang:

$$v = v(s) = \sqrt{\frac{mg}{k}\left(1 - e^{-\frac{2ks}{m}}\right)}$$

wobei m die Masse des Körpers, g die Erdbeschleunigung und k der Reibungskoeffizient (mit $k > 0$) ist. Wie lautet dieses Fallgesetz im luftleeren Raum?

Ejercicio 11.12.5 ("Angeschnittener" Wechselstrom)**.** Ein "angeschnittener" Wechselstrom, dessen Zeitabhängigkeit im Periodenintervall $0 \leqslant t \leqslant T$ durch die Gleichung

$$i(t) = \begin{cases} \hat{i} \cdot \cos(\omega_0 t) & 0 \leqslant t \leqslant \frac{T}{4} \\ 0 & \frac{T}{4} \leqslant t \leqslant T \end{cases}$$

beschrieben wird (Kreisfrequenz $\omega_0 = \frac{2\pi}{T}$). Wie lautet die Fourier-Zerlegung dieser Funktion in reeller Darstellung?

11.13. Gewönliche Differenzialgleichungen erster Ordnung

Ejercicio 11.13.1 (Differenzialgleichung mit Sinusfunktion)**.** Bestimmen Sie die Lösung des folgenden Anfangswertproblems:

$$\frac{dy}{dx} = (y+1) \cdot \sin x, \quad y\left(\pi/2\right) = 4.$$

Ejercicio 11.13.2 (Differenzialgleichung mit "besonderer Lösung")**.** Bestimmen Sie die Lösung des folgenden Anfangswertproblems:

$$\frac{dy}{dx} = 1 - y^2, \quad y(0) = 0.$$

Ejercicio 11.13.3 (Variation der Konstanten)**.** Bestimmen Sie die allgemeine Lösung der folgenden inhomogenen Differenzialgleichung mit der Methode der Variation der Konstanten:

$$y' + 4xy = 4x \cdot e^{-2x^2}.$$

Ejercicio 11.13.4 (Lineare Differentialgleichung mit konstanten Koeffizienten)**.** Bestimmen Sie die allgemeine Lösung der folgenden Differenzialgleichung durch "Aufsuchen einer partikulären Lösung":

$$\frac{dy}{dx} - 4y = e^{4x} + \cos(2x).$$